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Handbook of Investment Analysis, Portfolio Management, and Financial Derivatives Downloaded from www.worldscientific.com
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Chapter 1

Introduction to Investment Analysis, Portfolio Management, and Financial Derivatives

Cheng Few Lee

Contents

1.1 Introduction 2
1.2 Investment Analysis 2
1.2.1 Fundamental analysis 2
1.2.2 Technical analysis 4
1.2.3 Contrarian analysis 5
1.2.4 Dynamic asset allocation 5
1.3 Portfolio Analysis and Mutual Fund Management 5
1.4 Financial Derivatives 5
1.5 Overall Discussion of this Book 6
1.5.1 Chapter title classification 6
1.5.2 Keyword classification 7
1.6 Summary and Concluding Remarks 12
References 12

Abstract

The main purposes of this introduction chapter are (i) to give an overview of the following 109 papers, which discuss investment analysis, portfolio management, and financial derivatives; (ii) to classify these 109 chapters into nine topics; and (iii) to classify the keywords in terms of chapter numbers.

Keywords

Accounting beta • American option • ARIMA model • Basel Committee • Bayes factor • Bayesian test • Call option • CAPM • commodity futures market • Default • Delta • Disequilibrium effect • Earnings forecast • Executive compensation • Financial

stress • Fama–MacBeth regression • Financial technology • Fractional cointegration • GARCH model • Global financial crisis • Hedging performance • Implied variance • Investment strategy • January effect • Lagrangian multipliers • Long straddle • Machine learning • Markowitz modern portfolio theory • Momentum • Node clustering • Non-central chi-square distribution • Odd lot theory • Option bound • Poisson process • R Language • Short vertical (bear) spread • Simultaneous equation • Tail wag the dog • Tobin Q • Utility theory • Value-at-risk model • Weather derivatives

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1.1 Introduction

Investment analysis, portfolio management, and financial derivatives are important subjects for finance students, academicians, and practitioners. This handbook tries to present important papers in these three areas from theory, method, and application.

Investment analysis techniques can be classified into technical analysis, fundamental analysis, contrarian analysis, and dynamic asset allocation. There are different methodologies for portfolio analysis. They include optimization, minimization, and other methods which will be used to obtain the optimal weight of portfolio. Mutual fund topics can be regarded as one of the applications of portfolio analysis.

Financial derivatives include futures, options, swaps, and CDS. The valuations of future contracts and hedge ratio determination are important subjects. Future contracts are classified into individual stock future, index future, currency future, and commodity future. Options include the theory of evaluation, hedging, and their application in investment analysis and portfolio management.

The organization of this chapter is as follows. Section 1.2 will discuss investment analysis. Section 1.3 will discuss portfolio analysis. Section 1.4 will discuss financial derivatives. Section 1.5 will be an overall discussion of the book. A summary and concluding remarks will be presented in Section 1.6. Finally, the appendix will present the abstracts and keywords for Chapters 2 through 109.

1.2 Investment Analysis

1.2.1 *Fundamental analysis*

The job's main purpose of a security analyst is to determine the value of different securities. If a security's estimated value is above its market price, the security analyst will recommend buying the stock; if the value is below the market price, the security should be sold before its price drops. Underpriced

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stocks are purchased until their price is bid up to equal their value; overpriced stocks are sold, driving their price down until it equals their value.

There are two schools of thought as to how one determines an overpriced or underpriced security. **Fundamental analysis** (the fundamentalist school) studies the fundamental facts affecting a stock's value. Fundamental analysts delve into companies' earnings, their management, earnings forecasts, the firm's competition, market conditions, and many other business and economic factors. Alternatively, technical analysis can be used to determine an overpriced or underpriced security by studying the way security prices behave over time. Generally, this kind of analysis uses the moving average technique of other charting techniques.

The macro approach to fundamental analysis first emphasizes the analysis of the aggregate economy and market, then industry analysis, and finally the examination of specific companies within the industry. Changes in the national economy and credit conditions are associated with changes in interest rates, capitalization rates, and multipliers. Therefore, an aggregate economic and market analysis must be done in conjunction with determining the security's appropriate multiplier. A multiplier of 10 might be appropriate for normal economic conditions, whereas a multiple of 15 might be more realistic in an environment of rapid economic growth and prosperity. Overall, accounting information, economic conditions, and market information are three important pieces of information for security analysis.

All of the fundamentalist's research is based upon some valuation model. The analyst prepares his or her estimate of the intrinsic value per share at time 0, P_{i0}, by multiplying the ith stock's normalized earnings per share at time 0, E_{i0} times the share's earnings multiplier, m_{it}:

$$P_{i0} = E_{i0} m_{it} \tag{1.1}$$

Much of the fundamental analyst's work centers on determining the appropriate capitalization rate or equivalently the appropriate multiplier to use in valuing a particular security's income. This encompasses the micro approach to estimating future values for the stock market. It involves using a two-step approach: (1) estimating the expected earnings for some market-indicator series (Dow Jones Industrial Average or Standard & Poor's Industrial Index) or some stock and (2) estimating the expected earnings multiplier for the market series or stock. The main factors that must be considered in determining the correct multiplier are (1) the risk of the security, (2) the growth rate of the dividend stream, (3) the duration of any expected growth, and (4) the dividend payout ratio.

In determining the P/E ratio to use in valuing a firm's securities, three factors must be estimated: (1) the capitalization rate, (2) the dividend

growth rate, and (3) the dividend payout ratio. Algebraically,

$$\frac{P_0}{E} = \frac{d_1}{k-g} \quad \text{or} \quad \frac{P_0}{E} = \frac{d_1}{k-g} = \frac{d_0(1+g)}{k-g}, \tag{1.2}$$

where d_1/E is the dividend payout ratio, K is the capitalization rate, and g is the expected growth of dividends.

Given this equation, a positive relationship is expected between the earnings multiplier and the dividend payout, and with the growth rate of dividends, all things being equal. Alternatively, there should be a negative relationship between the earnings multiplier and the capitalization rate.

For further information about fundamental analysis, the reader can refer to Bodie *et al.* (2021), Lee *et al.* (2013, 2022), Lee and Lee (2017, 2022), and Ross *et al.* (2019).

1.2.2 *Technical analysis*

Technical analysts, rather than working through a large amount of fundamental information about an investment, such as company earnings, competitive products, and forthcoming legislation, searched for a summary of all facts by studying the way security prices behave historically. Over the past decades, technical analysts have focused their attention almost totally on charts of security market prices and related summary statistics.

Technical analysis is based on the widely accepted premise that security prices are determined by the supply of and the demand for securities. Typically, technical analysts record historical financial data on charts, study these charts in an effort to find meaningful patterns, and use these patterns to predict future prices. Technical analysts believe that past patterns of market action will recur in the future and that past patterns can be used for predictive purposes.

Rather than trying to evaluate the intrinsic value of a security, the technical analysts seek to estimate security prices; that is, they try to forecast short-run shifts in supply and demand that will affect the market price of one or more securities. Some of the tools used by chartists to measure supply and demand and to forecast security prices are the Dow theory chart, odd lot theory, confidence index, breadth-of-market indicators, relative strength analysis, and trading volume data.

For further information about technical analysis, the reader can refer to Bodie *et al.* (2021) and Lee *et al.* (2013).

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1.2.3 *Contrarian analysis*

A contrarian analyst is a person who opposes or rejects popular opinion, especially in stock exchange dealing. "It has become fashionable to be a stock-market contrarian." A contrarian analyst, in general, tries to select stocks which are not very popular with most investors, however, these stocks are fundamentally sound.

1.2.4 *Dynamic asset allocation*

In investment analysis, dynamic asset allocation is first to use both fundamental and technical analyses to choose stock and then use either option or future to hedge their investment. Mutual fund and hedge fund are frequently used dynamic asset allocation techniques for investment decisions.

For further information about dynamic asset allocation, the reader can refer to Lee *et al.* (2010, 2013, 2019), Bodie *et al.* (2021), and Lee and Lee (2015).

1.3 Portfolio Analysis and Mutual Fund Management

There are different methodologies for portfolio analysis. They include optimization, minimization, and other methods which will be used to obtain the optimal weight of portfolio. Mutual fund topics can be regarded as one of the applications of portfolio analysis. The reader can refer to Elton *et al.* (2013), Francis and Kim (2013), and Lee *et al.* (2013) for further details about this method.

Alternative mutual fund classification and their evaluation can also be found in these books. This handbook has covered this topic in detail.

1.4 Financial Derivatives

Financial derivatives include futures, options, swaps, and CDS. The valuations of future contracts and hedge ratio determination are important subjects. Future contracts are classified into individual stock future, index future, currency future, and commodity future. Options include the theory of evaluation, hedging, and their application in investment analysis and portfolio management. Lee *et al.* (2013, 2023), Bodie *et al.* (2021), and Hull (2018) have discussed this issue in detail. This handbook has extensively discussed the theoretical and methodological application of these topics in detail.

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1.5 Overall Discussion of this Book

In this section, we first classify 109 papers (Chapters 2–110) in accordance with subject content into nine groups and then we list the keywords with their associated chapter. From this section, the reader can get the overall of the handbook and have a better idea of how to use this handbook.

1.5.1 *Chapter title classification*

Based on the chapter titles, we classify 111 chapters into the following 9 topics:

(i) Financial Accounting and Auditing (2, 26, 30, 43, 44, 58, 70, 73, 82, 102)

These papers essentially show how accounting information and auditing information can be used to perform fundamental investment analysis.

(ii) Mutual Funds and Hedge Funds (22, 23, 37, 80, 94)

These papers discuss the fundamental issues of both mutual funds and hedge funds and how their performances can be evaluated and ranked.

(iii) Asset Pricing (7, 10, 20, 29, 88, 100)

These papers discuss theories and methodologies and how they can be used to perform fundamental or technical security analysis. Both equilibrium and disequilibrium models are explored in some detail.

(iv) Options (3, 5, 15, 27, 32, 47, 68, 69, 75, 77, 84, 86, 97, 104, 106, 107)

These papers discuss theories, methodologies, and applications of options. Theoretically we discuss how options instruments can be evaluated. Option strategies are discussed in some detail. Some of these papers also show how Excel program can be used to evaluate options and estimate implied volatility.

(v) Portfolio Analysis (4, 16, 19, 21, 24, 48, 56, 79, 85, 89, 90, 91, 93, 103)

These chapters discuss theories, methodologies, and applications of portfolio analysis. Theoretically, these papers cover both static and dynamic portfolio analyses. Besides academic methodology in portfolio analysis, the dynamic portfolio analysis developed by the industry is also explored.

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(vi) Risk Management (35, 41, 65, 81, 95, 98, 110)

These chapters discuss the issue of predicting default for banking credit, credit card, and other issues related to risk management. Some of these papers also discuss how corporate risk can be managed.

(vii) Investment Analysis, Momentum Analysis, and Behavior Finance (6, 9, 11, 12, 13, 14, 17, 18, 25, 28, 34, 36, 39, 42, 46, 49, 50, 53, 54, 55, 57, 59, 61, 62, 63, 64, 66, 67, 72, 76, 78, 87, 92, 96, 101, 105)

These papers discuss technical, fundamental, and momentum analyses. Different kinds of financial instruments are evaluated and analyzed in detail. International investment analysis is also covered in some detail.

(viii) Futures, Index Futures, and CDS (8, 33, 38, 45, 51, 52, 60, 74, 83)

These papers discuss financial futures, commodity futures, and CDS and their application to risk management for portfolio management. Alternative hedge ratio estimation method for futures is also explored.

(ix) Financial Econometrics, Statistics, and Technology (31, 40, 71, 99, 108, 109)

These papers show how econometrics, mathematics, statistics, and computer technology can be used to study investment analysis, portfolio analysis, and risk management.

Overall, these 109 chapters cover investment analysis, portfolio management, and financial derivatives in theory, method, and application. We believe this will be very useful for both academicians and practitioners.

1.5.2 *Keyword classification*

In this section, we list the keywords along with the chapter number it is associated with, then the reader can find the chapter where the keyword is located. This will help the reader more efficiently utilize the handbook.

Accounting analysis (76, 87); accounting anomalies (105); accounting beta (88); accounting cash flow (92); accounting information (82); accruals (26); accruals-based earnings management (73); adjustments (76, 87); American option (27, 77, 84); analyst heterogeneity (2); analysts' forecast dispersion (61); approximation approach (104); ARIMA models (94); ARIMA-GARCH model (17); asset allocation (87, 89); asset efficiency ratios (58); asset pricing (28, 54, 70, 83, 96); B-S

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model (32); balance sheet (70); bank (81); banking lending rate (56); banks (35); Basel Committee (44); Bayes factor (93); Bayesian test (93); behavioral biases (54); Benjamin Graham (64); beta coefficient (50, 56, 90); beta coefficients (29); betting against beta (48); bias of the estimated betas (108); binomial option pricing model (15, 69, 84, 101); Bitcoin (25); bivariate-normal distribution (97); Black–Litterman model (64, 85); Black–Scholes model (3, 69, 84, 97, 103); Black–Scholes option pricing model (15, 101); Black's model (97); Bollinger Bands (46); bond failure rates (98); bond rating (98); bond rating agencies (98); bond strategies (79); bonds (100); BOS ratio (14); business risk (56); calendar (time) spreads (107); call option (84); capital and finance leases (30); capital cash flow (92); capital mobility (6); capital requirements (35); capital structure (11, 12); CAPM (88); carbon emissions (55); Carhart four-factor model (55); cash (70); Cathie Wood (64); causality (8); CDS Big Bang (74); CDS-bond basis (74); CEV model (104); CFA (80); characteristic-based model (65); China's stock markets (42, 49); Chinese IPO (11); circular economy (66); climate change (55); co-dependence (9); coefficient determination (29); coinsurance (26); collar (107); combination forecasting model (17); combined investment strategy (14); commodities (8); commodity diversifier (21); commodity futures market (52); community detection (99); comovement (54); component analysis (94); composite forecasting (88, 94); compounded Poisson process (40); computational modeling (71); computational science (71); computer intensive methods (71); concave utility function (89); conditional multivariate F test (93); conditional value-at-risk model (103); conditional value-at-risk (4); confidence index (94); confirmatory factor analysis (CFA) (12); constant elasticity of variance model (106); constant elasticity of variance (CEV) process (86); constrained Kalman filter method (62); contagious effect (16); continuous wavelet analysis (21); conventional approach (45); copula function (16); Corporate Social Responsibility (RSC) (66); correlated defaults (16); cost of equity (11); cost of equity capital (41); covariance (90); covered call (107); COVID-19 (9); Cox process (32); credit default swaps (83); credit risk (44, 83); cross-country linkage (51); cross-quantilogram correlation (9); cross-sectional returns (59); CRSP value-weighted index (93); cryptocurrencies (25, 87); CSR reports (66); current earnings (109); current EPS (109); debt cash flow (92); default (44); delta (Δ) (47); demand function (20); derivatives (35); determinants of firm value (76, 87); diffusion process (40); discounted cash flow (92); Discounted Cash Flows (DCF) (63); disequilibrium effect (20); disequilibrium model (20); diversification (26, 48, 52, 56); dividend behavior models (109); dividend yield (97); Dow theory (94); dummy variable regression model (39); duration (79); dynamic factor model (16); dynamic hedging (95); dynamic market model (29); earnings

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(96); earnings forecast (2); earnings forecasts (102); earnings quality (26); earnings surprises (18); earnings volatility (26); economic policy uncertainty (9, 28); efficiency (43); EGARCH (49); EGARCH model (85); empirical performance (86); endogeneity (108); equities (100); equity cash flow (92); equity funding supply (11); equity market timing (11); equity premium (10); equity valuation (73); error correction approach (45); error correction model (17, 29); European call/put (69); European options (27, 84); European sovereign debt crisis (60); Eurozone debt crisis (6); event extraction (78); *ex ante* prediction (40); *ex post* filter (40); *ex post* Sharpe ratio (93); Excel (69); Excel program (84); Excel VBA (84); excessive risk-taking (37); exchange option (32); executive compensation (110); Expected Credit Loss (ECL) (44); expected returns (102); factor model (65); factor models (64, 96, 102); Fama–MacBeth regression (108); family firms (57); feedback trading (23); Feldstein–Horioka puzzle (6); financial accounting (70); financial analysis (76, 87); financial crisis (72); financial derivatives (75); financial econometrics (31); financial flexibility (72); financial leverage ratios (58); financial markets (75); financial mathematics (31); financial options (3); financial risk (56); financial risk hedging (41); financial statement analysis (14); financial statistics (31); financial stress (9); financial technology (31); financial z-score (58); finite difference method of the SV model (86); Fintech (25); firm valuation (82); firm value (55); forecast aggregation (2); forecasting (8); founder firms (57); founder horizon (57); founder influence (57); founder tenure (57); four-factor model (7); fractional cointegration (34); free cash flow (92); freight derivative (75); fund competition (37); fund flow (80); fund management (80); fund performance (80); fundamental analysis (14, 63, 94); funding liquidity (74); future option (97); fuzzy multiple criteria decision-making (FMCDM) (22); Gamma (Γ) (47); GARCH method (38); GARCH model (85); GARCH-jump mixture model (40); GARJI model (40); generalized method of moments (24); generalized singular value decomposition (GSVD) (62); genetic programming (59); global financial crisis (6); global financial crisis (81); gold (21); goodness of fit (40); great investors (64); greenhouse gas (55); greenwashing (66); GRS test (96); Hawkes process (16); hedge funds (80); hedge ratio (33, 38, 45, 51); hedging (3, 47, 75, 87); hedging performance (45); herding (23, 100); heterogeneity (98); hierarchical system (22); historical volatility (68); home bias (19); homogeneity (98); IAS 39 (44); IBEX 35 Index (66); IFRS 9 (44); implied variance (104); implied volatility (35, 68); implied volatility spread (10, 13); incomplete market (27); Incurred Credit Loss (ICL) (44); indifference curve (89); industry 4.0 (25); information asymmetry (41); information fusion (78); informativeness of financial markets (5); inside debt (110); insider trading (72); institutional trading (23); instrumental variable (108); interest rate sensitivity

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(61); interest rate anticipation swap (79); intermarket spread swap (79); international diversification (19); international equity markets (24); intervention (17); intrinsic value (63); inverse Fourier transform (32); investment analysis (4); investment constraints (19); investment horizons (21); investment portfolio formation (76, 87); investment potential (43); investment strategies (64); investors' disagreement (61); James Simons (64); January effect (39); Jensen performance measure (94); joint normality hypothesis (33); kurtosis (31); Lagrangian calculus maximization (90); Lagrangian multipliers (91); Lagrangian objective function (89); lead–lag relation (68); lead–lag relationship (52); legal environment (24); leverage (35); leverage effect (40); linear programming (27, 90); linear regression (46); linear utility function (89); link-pattern communities (99); liquidity (43, 65, 81, 83, 96); liquidity ratios (58); liquidity timing (80); LISREL (12); log-normal distribution (77); log-normal distribution method (15, 101); long call (107); long put (107); long straddle (107); long straddle (107); long vertical (bull) spread (107); long-run multiplier (29); long-term investments (76, 87); long-term payout ratio (109); low cross-sectional correlation (48); low-variability anomaly (48); machine learning (31, 53, 59); MAPE (17); market beta (88); market efficiency (82); market model (50, 88, 90); market risk (55); market structure (83); market timing (80); market value ratios (58); Markov chains (98); Markowitz modern portfolio theory (85); MATLAB approach (104); maturity (79); maturity mismatches (61); maximum likelihood estimator (20); mean–variance efficiency (93); minimum variance hedge ratio (38); mispricing (65); modern portfolio theory (64); momentum (67, 96); momentum factor (13, 105); momentum strategies (14, 18); Monte Carlo simulation (16, 71); moving average (94); multiperiod dynamic CAPM (20); multiple-index model (90); multivariate log-normal distribution (77); multivariate normal distribution (77); mutual fund (37); mutual funds (22, 23); N-Period OPM (84); natural language processing (53); news headlines (53); node clustering (99); non-additive fuzzy integral (22); non-financial information (66); non-parametric (27); non-systematic risk (88); non-normality (8); non-central chi-square distribution (106); non-interest income share (81); normal distribution (77); numerical experiment (86); odd lot theory (94); off-balance sheet leases (30); oil (8, 9); omega model (4, 103); On-air stock reports (36); on-balance sheet leases (30); one-period OPM (84); operating (30); operating performance (57); optimal weights (56); optimum mean–variance hedge ratio (38); option (68); option bound (77); option bounds (27); option price (13); option pricing (106); option pricing model (86); options (35); order submission behavior (36); oscillators (46); OTC derivatives (83); outlier

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detection (17); overconfidence (110); P2P lending (25); pan-Chinese markets (19); panel unit root tests (6); partial adjustment coefficient (109); past stock returns (13); payout policy (72); permanent earnings (109); permanent EPS (109); plastic industry (66); Poisson process (32); policy ratios (58); portfolio choice (48); portfolio rebalancing (4); portfolio selection (103); post-earnings announcement drift (18, 105); predictability (59); prediction (10); predictive power (7); price pressure (13); prices (8, 36); probability of default (98); profitability (43); profitability ratios (58); prospective analysis (76, 87); protective put (107); Provision for loan losses (44); pure martingale and joint normality hypotheses (33); pure martingale hypothesis (33); pure yield pickup swap (79); put option (84); put options (95); put–call parity (107); quantitative analysis (78); R language (38); R language programming (69); R&D investment (110); rating recalibration (54); ratio analysis (43); real earnings management (73); real estate (100); real-time responses (36); realized volatility (35); residual operating income valuation model (76, 87); return comovement (48); return momentum effect (105); return predictability (54); return reversals (28); revenue surprises (18); rho (ρ) (47); risk (65); risk aversion (89); risk-free rate (91); risk-taking incentive (110); RMSE (17); robust index tracking (62); sample filtering (67); SAS language programming (69); saving–investment correlation (6); securities accounts (11); semi-parametric (27); sentiment analysis (53); sentiment analysis (78); shape parameter (40); Sharpe hedge ratio (38); Sharpe performance measure (56, 91, 94); short call (107); short put (107); short sales allowed (91); short sales not allowed (91); short selling (89, 103); short straddle (107); short vertical (bear) spread (107); short-term debt (70); simulation (71); simultaneous equation (51); single-index model (90); size variable (108); skewness (31, 67); social network (78); socially responsible investment (SRI) (66); solvency (43); sovereign debt (83); sovereign risk ratings (60); spanning tests (95); specification analysis (109); spillover (42); static factor model (16); stationary (34); statistical power of test (108); stochastic calculus (15, 101); stochastic dominance (27); stochastic volatility option pricing model (86); stock correlation (78); stock index futures (95); stock indices (69, 97); stock market (52); stock market momentum (13); stock market seasonality (39); stock markets (9); stock prediction (78); stock repurchase (72); stock return (55); stock return directions (53); stock volatility (41); stop-loss orders (95); strategy (57); strategy analysis (76, 87); structural equation modeling (SEM) (12); style investing (54); style momentum (54); substitution swap (79); supply function (20); support and resistance (46); sustainability reports (66); swapping (79); synthetic option (84, 95); systematic and idiosyncratic risk

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(110); systematic risk (54, 88); t statistics (29); tail wag the dog (95); technical analysis (14, 46, 59, 94); TED spread (34); temporal aggregation (8, 29, 45); textual analysis (53); the abnormal earnings growth model (82); the investor's views (85); the residual income valuation model (82); theta (Θ) (47); three-stage estimated returns (7); three-stage least squares (51); Tobin Q (58); total risk (88); tournament (37); trade credit (70); trading volume (14, 42, 61, 94); transaction costs (4, 103); transfer function model (17); transition probability matrix (98); transparency (57); trend lines (46); Treynor performance measure (91, 94); triangular fuzzy numbers (22); turning point method (17); two-period OPM (84); two-stage least squares (2SLS) estimator (20); UK firms (41); utility function (89); utility theory (89); value-at-risk model (103); value investing (63); value line enigma (50); value line ranking (50); value relevance (55, 82); variance gamma model (40); Vega (ν) (47); VG NGARCH model (40); Vietnam (25); virtual currency (25); VIX as a measure of market nervousness/uncertainty (5); volatility (3, 49, 67); volatility clustering (40, 85); volatility risk measure (35); volatility risk premia (35); volatility skew (35); volatility spillover effects (60); volatility spread (35); volatility timing (80); volume (36); Warren Buffett (64); weather derivatives (75); and wholesale funding (81).

1.6 Summary and Concluding Remarks

In this chapter, we have discussed the issues of investment analysis, portfolio management, and financial derivatives. Then we discussed the structure of this book, which classifies 109 chapters into nine topics. In addition, we have classified the keywords in terms of the chapter numbers. In the appendix, we present the abstracts and keywords.

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Appendix. Abstracts and Keywords for Chapters 1 to 112

Chapter 1: Introduction

The main purposes of this introduction chapter are (i) to give an overview of the following 109 papers, which discuss investment analysis, portfolio management, and financial derivatives; (ii) to classify these 109 chapters into nine topics; and (iii) to classify the keywords in terms of chapter numbers.

Keywords
Accounting beta, American option, ARIMA model, Basel Committee, Bayes factor, Bayesian test, call option, CAPM, commodity futures market, default, delta, disequilibrium effect, earnings forecast, executive compensation, financial stress, Fama–MacBeth regression, financial technology, fractional cointegration, GARCH model, global financial crisis, hedging performance, implied variance, investment strategy, January effect, Lagrangian multipliers, long straddle, machine learning, Markowitz modern portfolio theory, momentum, node clustering, non-central chi-square distribution, odd lot theory, option bound, Poisson process, R language, short vertical (bear) spread, simultaneous equation, tail wag the dog, Tobin Q, utility theory, value-at-risk model, weather derivatives

Chapter 2: Analyst characteristics-based consensus forecasts

Consensus earnings forecasts matter to investment practitioners in forming profitable investment portfolios and matter to researchers in studying how earnings inform the market. This study revisits the issue of considering analyst heterogeneity in forming better analyst consensus earnings forecasts (Clement, 1999; Clement and Tse, 2003; Brown and Mohd, 2003).

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Based on quarterly data for US firms from 1994 to 2017, the study finds that characteristics-based consensus forecasts outperform the simple mean consensus to predict abnormal returns in both the three-day window around and the two-month drift window after earnings announcements. They perform as well as the median consensus does to predict abnormal returns in the three-day window around earnings announcements but outperform the median consensus in the drift window. Investors may find these findings relevant to their investment decisions, and researchers may find them relevant to the studies of earnings informativeness.

Keywords
Earnings forecast, analyst heterogeneity, forecast aggregation

Chapter 3: Models of option pricing

Starting from humble beginnings, the use of financial options has substantially increased as an important financial tool for both speculation and hedging over the last 50 years. This chapter discusses both the theoretical and practical applications of financial options and related models. While the content is somewhat technical, we provide illustrations of their applications in simple settings. We address particular stylized features of option pricing models.

Keywords
Financial options, Black–Scholes model, volatility, hedging

Chapter 4: Realized diversification benefits of risk portfolio models

In this chapter, we apply various portfolio models to rebalance portfolios and further analyze their realized performance, including mean–variance (MV), conditional value-at-risk (CVaR), and omega models. To ensure feasibility, we consider the transaction costs and the optimization of short-selling weights. The empirical results using the daily returns of international funds across 21 countries over 20 years show that the risky portfolios realize higher performance than a naïve diversification, particularly the CVaR model with a confidence level of 95%, despite their higher trading costs. The CVaR model, mainly focusing on controlling loss, yields higher performance than those that are based on trade-off between return and volatility, such as the mean–variance and omega models. The omega model, however, generates lower downside risk. The superiority of risky portfolios over the equally

weighted diversification varies intertemporally across various models. The excess returns of risky portfolios over the equally weighted diversification are larger when the market is volatile, such as the periods of the subprime mortgage financial crisis and the 2020 COVID-19 recession.

Keywords
Investment analysis, portfolio rebalancing, conditional value-at-risk, omega model, transaction costs

Chapter 5: VIX implied volatility as a time-invariant, stationary assessor of market nervousness/uncertainty

Financial markets serve numerous roles, amongst them of course is the uncoerced exchange of securities. In addition to that role, they serve a very useful function of conveying to market observers information about the future, the challenge being our ability to elicit and interpret that information.

This paper addresses that latter function regarding the option markets which provide the value for the VIX 30-day implied volatility on the S&P 500 market index. It is demonstrated that the peak value of VIX during Persian Gulf I, 1990/1991, and Persian Gulf II, 2003, was nearly identical.

Keywords
Informativeness of financial markets, VIX as a measure of market nervousness/uncertainty

Chapter 6: Investment and saving in the Eurozone and the European Union: another look at Feldstein–Horioka

In this chapter, we employ data from a panel consisting of 14 EU member countries over the period 1970–2020 to examine the validity of the Feldstein–Horioka puzzle (F–H puzzle). Despite many criticisms, the F–H saving—investment correlation hypothesis is still used in the literature to infer the degree of capital mobility among countries. To this end, we apply a battery of panel unit root and co-integration tests. The finding of the presence of co-integration of the savings and investment ratios and the observed magnitude of the estimated average savings–retention coefficient for the panel reveals that for this panel of EU member countries, the F–H puzzle is not valid and the long-run international solvency condition is maintained in most of these countries. The observed low savings–retention coefficients for these countries imply a moderate degree of capital mobility and the absence of the Feldstein–Horioka puzzle. This finding of the prevalence of a moderate

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degree of capital mobility is consistent with the macroeconomic experience of these countries during the period under investigation.

Keywords
Capital mobility, Feldstein–Horioka puzzle, saving–investment correlation, panel unit root tests, global financial crisis, Eurozone debt crisis

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Chapter 7: A three-stage procedure for predicting stock returns

In this chapter, we apply a three-stage approach using an intermediate classification period between the estimation and test period. In the intermediate period, we stratify individual firms into deciles based on the predictive power of the Carhart four-factor model, measured by the out-of-sample R-squared prediction in this period. Our motive for this stratification is that firms with poor out-of-sample predictive power of the estimated model are likely to suffer from coefficient instability and that these instabilities will result in a mismeasurement of expected returns in the test period. The empirical results show that lower predictive power deciles have larger averaged absolute changes of estimated coefficients, which is our proxy for coefficient instability, and lower averaged out-of-sample R-squared in the test period.

Keywords
Four-factor model, predictive power, three-stage estimated returns

Chapter 8: Temporal aggregation and the estimation of reverse regressions for commodities market models

The main purpose of this article is to empirically demonstrate the effects of temporal aggregation when applying reverse regression models hypothesizing that spot prices today help predict forward rates in the future. This paper essentially reviews results from earlier research indicating that time-series aggregation will most certainly influence standard errors on parameter estimates. Standard errors are likely to increase with aggregation. The relationships between futures prices and spot oil prices are analyzed along with the importance of the effects of temporal aggregation and alternative model specifications for understanding empirical relationships between the markets. Model specification and time-series aggregation over daily, weekly, and monthly aggregations confirm evidence that estimated standard errors are likely to increase with aggregation and t-ratios change as well. While goodness-of-fit measures might increase with aggregation, forecast accuracy

with macro-level aggregation might deteriorate owing to information loss due to the averaging of observations associated with underlying micro-structures.

Keywords
Commodities, prices, temporal aggregation, oil, causality, forecasting, non-normality

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Chapter 9: Correlation and dependence between oil prices, stock returns, policy uncertainty, and financial stress during COVID-19 pandemic: New evidence from a multicountry analysis using cross-quantilogram method

In this chapter, we examine risk spillover between the returns series of oil and stock prices of worst-affected countries due to the COVID-19 outbreak in unconditional and conditional frameworks, where the relationship was conditioned upon the US economic policy uncertainty and financial stress indices. Specifically, we used three different measures of oil prices, namely, WTI, OPEC, and Dubai oil prices. We also examined the risk spillover from US and Chinese stock markets to stock markets of affected countries, such as the UK, France, Germany, Italy, Spain, Switzerland, and Turkey for the time period from 31st December 2019 to 22nd April 2020. Our results provide evidence that during the COVID-19 outbreak, Dubai and OPEC oil prices have had a strong positive effect on stock price when both of them are at their lower quantiles, which suggests that during extreme markets conditions, oil price affects stock price. Furthermore, evidence of the directional predictability from stock returns of US/China to all other stock returns shows a positive predictability from the US to France, Germany, Italy, Spain, Switzerland, and the UK at lower quantiles. Last but not least, when the relationships were conditioned by the policy uncertainty and financial stress, evidence of directional predictability become stronger and spread to more quantiles, suggesting that the interrelationships between oil price and stock price returns and between stock price returns of US/China to all other stock price returns were not driven by the systemic risk but rather by uncertainties during the COVID-19 outbreak.

Keywords
Co-dependence, oil, stock markets, cross-quantilogram correlation, COVID-19, economic policy uncertainty, financial stress

Chapter 10: Predicting the equity premium with the implied volatility spread

We show that the call–put implied volatility spread (IVS) outperforms many well-known predictors of the U.S. equity premium at return horizons up to

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six months over the period from 1996:1 to 2017:12. The predictive ability of the IVS is unrelated to the dividend yield and is useful in explaining the cross-section of returns. Decomposing the IVS, we find that the longer run predictive ability of the IVS operates primarily through a cash flow channel. We also find that the IVS is significantly related to indicators of aggregate market direction and expected market conditions. Our results are consistent with the IVS reflecting market sentiment as well as information about informed trading.

Keywords
Implied volatility spread, equity premium, prediction

Chapter 11: Does equity market timing have a persistent impact on capital structure? Evidence from China

This paper uses the change in individual securities accounts as a measure of equity funding supply to examine whether the persistent timing effect on capital structure exists for the Chinese equity market. This new equity timing measure avoids previous criticisms over a timing measure not being independent of a firm's characteristics of capital structure. Our empirical results show that this new measure is an effective market timing variable for issuing equity in the Chinese equity market and that a persistent effect of equity market timing on firm capital structure exists for more than seven years. This paper offers evidence that the market conditions of equity funding supply play an important role in corporate financing decisions in China.

Keywords
Cost of equity, equity market timing, capital structure, equity funding supply, Chinese IPO, securities accounts

Chapter 12: The joint determinants of capital structure and stock rate of return: a LISREL model approach

We develop a simultaneous determination model of capital structure and stock returns. Specifically, we incorporate the managerial investment autonomy theory into the structural equation modeling with confirmatory factor analysis to jointly determine the capital structure and stock return. Besides attributes introduced in previous studies, we introduce indicators affecting a firm's financing decision, such as managerial entrenchment, macroeconomic factors, government financial policy, and pricing factors. Empirical results show that stock returns, asset structure, growth, industry classification,

uniqueness, volatility, financial rating, profitability, government financial policy, and managerial entrenchment are major factors of the capital structure.

Keywords
LISREL, structural equation modeling (SEM), confirmatory factor analysis (CFA), capital structure

Chapter 13: Alternative methods for estimating firm's growth rate: Update and extension

The growth rate plays an important role in determining a firm's asset and equity values. The premier model used to price equity is the basic dividend growth model of Gordon and Shapiro (1956). Nevertheless, the basic assumptions of the growth rate estimation model are less well understood. In this chapter, we demonstrate that the model makes strong assumptions regarding the financing mix of the firm. In addition, we discuss various estimation methods of the firms' growth rate. We demonstrate that the arithmetic average method is very sensitive to extreme observations, and the regression methods yield similar but somewhat smaller estimates of the growth rate compared to the compound-sum method. Interestingly, the ex post forecast shows that arithmetic average and compound-sum methods (continuous regression) yield the best (worst) performance with respect to estimating a firm's future dividend growth rate. Firm characteristics, like size, book-to market ratio, and systematic risk, have a significant influence on the forecast errors of dividend and sales growth rate estimation.

Keywords
Arithmetic average method, compound-sum method, continuous regression method, discount cash flow model, discrete regression method, dividend growth model, geometric average method, Gordon's growth model, internal growth model, sustainable growth model

Chapter 14: Technical, fundamental, and combined information for separating winners from losers

This study examines how fundamental accounting information can be used to supplement technical information to separate momentum winners from losers. We first introduce a ratio of liquidity buy volume to liquidity sell volume (BOS ratio) to proxy the level of information asymmetry for stocks and show that the BOS momentum strategy can enhance the profits of momentum strategy. We further propose a unified framework, produced by incorporating two fundamental indicators — the FSCORE (Piotroski,

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2000) and the GSCORE (Mohanram, 2005) — into momentum strategy. The empirical results show that the combined investment strategy includes stocks with a larger information content that the market cannot reflect in time, and, therefore, the combined investment strategy outperforms momentum strategy by generating significantly higher returns.

Keywords
BOS ratio, combined investment strategy, financial statement analysis, fundamental analysis, momentum strategies, technical analysis, trading volume

Chapter 15: Alternative methods to derive option pricing models: Review and comparison

The main purposes of this paper are (i) to review three alternative methods for deriving option pricing models (OPM), (ii) to discuss the relationship between binomial OPM and Black–Scholes OPM, (iii) to compare Cox *et al.* method and Rendleman and Bartter method for deriving Black–Scholes OPM, (iv) to discuss log-normal distribution method to derive Black–Scholes OPM, and (v) to show how the Black–Scholes model can be derived by stochastic calculus.

This paper shows that the main methodologies used to derive the Black–Scholes model are binomial distribution, log-normal distribution, and differential and integral calculus. If we assume risk neutrality, then we don't need stochastic calculus to derive the Black–Scholes model. However, the stochastic calculus approach for deriving the Black–Scholes model is still presented in Section 15.6. In sum, this paper can help statisticians and mathematicians understand how alternative methods can be used to derive the Black–Scholes option model.

Keywords
Black–Scholes option pricing model, binomial option pricing model, log-normal distribution method, stochastic calculus

Chapter 16: An assessment of copula functions approach in conjunction with factor model in portfolio credit risk management

In credit risk modeling, factor models, either static or dynamic, are often used to account for correlated defaults among a set of financial assets. Within the realm of factor models, default dependence is due to a set of common

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systematic risk factors. By coupling with a copula function, *e.g.*, the normal, t-, Clayton, Frank, and Gumbel copula function, an analytic formulation of the joint distribution of assets' default times can be derived. On the other hand, factor models fail to account for the contagion mechanism of defaults in which a firm's default risk increases due to their commercial or financial counterparties' defaults. This study considers the dynamic factor model of Duffee (1999) coupling with a Hawkes process, a class of counting processes allowing intensities to depend on the timing of previous events (Hawkes, 1971), for the contagious effect. Using the factor-contagious-effect model, Monte Carlo simulation is performed to generate default times of two hypothesized firms. It is demonstrated that as the contagious effect increases, the goodness of fit of the joint distribution of assets' default times based on copula functions decreases, which highlights the deficiency of the copula function approach.

Keywords
Static factor model, dynamic factor model, correlated defaults, copula function, contagious effect, Hawkes process, Monte Carlo simulation

Chapter 17: Forecast performance of the Taiwan weighted stock index: Update and expansion

This research introduces the following to establish a TAIEX prediction model: intervention analysis integrated into the ARIMA-GARCH model, ECM, intervention analysis integrated into the transfer function model, the simple average combination forecasting model, and the minimum error combination forecasting model. The results show that intervention analysis integrated into the transfer function model yields a more accurate prediction model than ECM and intervention analysis integrated into the ARIMA-GARCH model. The minimum error combination forecasting model can improve prediction accuracy much better than non-combination models and also maintain robustness. Intervention analysis integrated into the transfer function model shows that the TAIEX is affected by external factors, the INDU, the exchange rate, and the consumer price index; therefore, facing the different issues of the TAIEX, the government could pursue some macroeconomic policies to reach the goals of policies.

Keywords
ARIMA-GARCH model, transfer function model, intervention, combination forecasting model, outlier detection, error correction model, MAPE, RMSE, turning point method

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Chapter 18: Does revenue momentum drive or ride earnings or price momentum?

This paper examines the profits of revenue, earnings, and price momentum strategies in an attempt to understand investor reactions when facing multiple information on firm performance in various scenarios. We first offer evidence that there is no dominating momentum strategy among the revenue, earnings, and price momentums, suggesting that revenue surprises, earnings surprises, and prior returns each carry some exclusive unpriced information content. We next show that the profits of momentum driven by firm fundamental performance information (revenue or earnings) depend upon the accompanying firm market performance information (price), and vice versa. The robust monotonicity in multivariate momentum returns is consistent with the argument that the market does not only underestimate the individual information but also the joint implications of multiple information on firm performance, particularly when they point in the same direction. A three-way combined momentum strategy may offer monthly return as high as 1.44%. The information conveyed by revenue surprises and earnings surprises combined accounts for about 19% of price momentum effects, which finding adds to the large literature on tracing the sources of price momentum.

Keywords
Revenue surprises, earnings surprises, post-earnings announcement drift, momentum strategies

Chapter 19: Do investors still benefit from culturally home-biased diversification? An empirical study of China, Hong Kong, and Taiwan

Due to the continual economic integration and the accumulation of wealth in China, Hong Kong, and Taiwan since early 1990's, understanding portfolio strategies and the benefits of diversification for these countries is an indispensable element in managing global assets. Using weekly industry-level data, we analyze the culturally home-biased diversification and find that local investors still benefit from regional investments. The time-varying benefits of diversification exist even as the economies of this region have become increasingly integrated. Our analysis suggests that stricter weighting bounds reduce the economic values of diversification but enhance the feasibility of the optimal portfolio allocations. The larger benefits gained by Chinese investors

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suggest that international diversification is *more* advantageous to investors in emerging economies than those in developed markets. The robustness tests generate similar findings when we evaluate the out-of-sample effectiveness and the benefits of diversification under various parameter estimation windows.

Keywords
Pan-Chinese markets, investment constraints, home bias, international diversification

Chapter 20: Product market competition and real activities manipulations: Theory, implications, and applications

We investigate how a firm manipulates its real activities in production to meet the earnings target in product market competition against its product-market rivals. We show that the equilibratory way to reach the earnings target is to set a higher first-period output level, reaching a higher short-term profit level. However, once the expected level of demand uncertainty is high, a firm will exploit this effect on its output choice by taking a mixed strategy and raising its short-term output level. This result suggests that one should consider longer-horizon paths of variables to detect opportunistic real activities manipulation. Based on our results, we further argue that competitive strategy is an omitted variable in real activities manipulation estimation models and recommend that capacity utilization, which is related to a firm's output competitive decisions, should be included in the first-stage models of normal investment levels in Roychowdhury (2006) and Gunny (2010).

Keywords
Earnings target, real activities manipulation, Cournot competition, one-sided, incomplete information

Chapter 21: Gold in portfolio: A long-term or short-term diversifier?

The purpose of this chapter is to evaluate the role played by gold in a diversified portfolio comprised of bonds and stocks. The continuous wavelet transform analysis is applied to capture the correlation features between gold and other risky assets at a specific time horizon to determine whether gold should be included in a diversified portfolio. This chapter uses the U.S. stock,

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bond, and gold data from 1990 until 2020 to investigate the optimal weights of gold obtained from the minimum variance portfolio. Empirical findings suggest that little evidence support that gold acts as an efficient diversifier in traditional stock and bond portfolios. Gold typically has been a long-term diversifier in the traditional portfolio comprised of bonds and stocks only before the early 2000s and acts as a short-term diversifier in times of crisis periods. The significant drop in the long-term weight of gold indicates that gold losses much of its long-term role in the diversified portfolio. These findings are useful for portfolio managers to justify the gold's diversification benefits over different investment horizons.

Keywords
Gold, commodity diversifier, investment horizons, continuous wavelet analysis

Chapter 22: Fuzzy multicriteria decision-making for evaluating mutual fund strategies

Investors often need to evaluate the investment strategies according to their own subjective preferences in terms of numerical values based upon various criteria when making investment in mutual funds. This situation can be regarded as a fuzzy multiple criteria decision-making (MCDM) problem. The purpose of this study is to propose an alternative approach, fuzzy multiple criteria decision-making with fuzzy integral. This approach relaxes the independence assumption among criteria for the evaluation of the MCDM problems, which is oftentimes the basic assumption in applying a hierarchical system for evaluating the strategies of selecting the mutual funds' investment style. We also employ triangular fuzzy numbers to represent the decision makers' subjective preferences on the criteria, as well as for the criteria measurements to evaluate mutual funds' investment style. To achieve this objective, first, we employ factor analysis to extract four independent common factors from those criteria. Second, we construct the evaluation frame using a hierarchical system composed of the above four common factors with 16 evaluation criteria and then derive the relative weights with respect to the considered criteria. Third, the synthetic utility value corresponding to each mutual fund's investment style is aggregated by the fuzzy weights with fuzzy performance values. Finally, we compare with empirical data and find that the model of FMCDM predicts the rate of return very accurately in certain ranges of λ, hence the non-additive fuzzy integral technique is an effective method for evaluating mutual funds' strategy.

Keywords
Mutual funds, fuzzy multiple criteria decision-making (FMCDM), hierarchical system, non-additive fuzzy integral, triangular fuzzy numbers

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Chapter 23: Mutual fund herding and its impact on stock returns: Evidence from the Taiwan stock market

Using quarterly ownership data which identify identity codes of mutual funds in Taiwan, we investigate mutual fund herding and its impact on stock price. We show that mutual funds tend to follow their own steps in trading rather than follow trades made by other funds. More importantly, evidence of price continuation following mutual fund herd buying suggests that such herding is based on value-relevant information and is consistent with the investigative herding hypothesis. Alternatively, evidence of return reversal following mutual fund herd selling suggests that such herding is non-informational and is consistent with the characteristic herding hypothesis.

Keywords
Mutual funds, herding, feedback trading, institutional trading

Chapter 24: Stock return, risk, and legal environment around the world

This chapter shows how the legal environment in a country influences the performance and risk of stock across countries at different developmental stages and of various rules of jurisdiction. Using data of 4,916 stocks from 37 countries, our empirical findings confirm that equities in countries with English common law origin have higher risk premiums than those in civil law countries, particularly for countries of the French/Spanish code. The indicators representing high efficiency in the law system, low corruption, strong legal protection of investors' rights, and reliable political environment are associated with low risk and high performance. The various elements of legal procedural formalism, however, have differing effects on volatility and return.

Keywords
Legal environment, international equity markets, generalized method of moments

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Chapter 25: Further analysis of Bitcoin, Fintech, and P2P lending: Perspectives and recommendations from industry 4.0

In several countries in the world, Bitcoin and P2P lending has been accepted and developed strongly. This study aims to evaluate the suitability, pros, and cons of Bitcoin, Fintech, P2P lending, and its platform in emerging markets, such as Vietnam. The research used qualitative analysis combined with data collection method published, statistics, analysis, synthesis, comparison, to generate qualitative comments and discussion; evaluate results, the article analyzed and evaluated the impacts of Fintech, P2P lending, and Bitcoin and virtual currency on the society of Vietnam, both positive and negative sides. It was found that we need to improve regulations on Fintech and shadow banking to overcome the weaknesses of commercial banks, to reduce risks as many nations in the world accept it. Experiences of other countries, such as the United States, Japan, China, or the developed countries of the European Union, and consequences for the economy became a lesson for well as developing countries. Hence, we need to implement risk management plans to reduce technological and IT risks. Proper solutions and development orientation as well as risk management for Bitcoin and cryptocurrencies are suggested. Last but not least, the research was limited to the case of Vietnam; hence, we can expand research to other Asian countries or other emerging markets.

Keywords
Bitcoin, virtual currency, cryptocurrencies, Fintech, P2P lending, industry 4.0, Vietnam

Chapter 26: Earnings quality and the coinsurance effect

We argue that coinsurance among a firm's business units changes the properties of reported earnings through less volatile operations (financial synergies) and fewer estimation errors in the accrual process (accounting synergies). Consistent with a coinsurance effect, we find that diversified firms have on average higher earnings quality compared to industry-matched portfolios of focused firms. Specifically, diversification leads to more predictable earnings, superior mapping of accruals to cash flows, and lower absolute abnormal accruals. In addition, we find higher earnings quality for diversified firms with less correlated segment earnings and that the coinsurance effect is stronger for firms that operate in more volatile and uncertain environments.

We contribute by identifying the coinsurance effect of diversification as a new determinant of earnings quality. Our findings complement prior literature on agency-related disadvantages of diversification for earnings quality by highlighting coinsurance-related benefits of diversification for earnings quality.

Keywords
Coinsurance, accruals, earnings volatility, earnings quality, diversification

Chapter 27: Alternative methods for determining option bounds: A review and comparison

This paper first reviews alternative methods for determining option bounds. This method includes stochastic dominance, linear programming, semi-parametric method, and non-parametric method for European option. Then option bounds for American and Asian options are discussed. Finally, we discuss empirical applications in equities and equity indices, index futures, foreign exchange rates, and real options.

Keywords
Option bounds, stochastic dominance, linear programming, semi-parametric, non-parametric, incomplete market, American options, European options

Chapter 28: Economic policy uncertainty and short-term reversals

This study finds that short-term reversals become more profound in the current month when economic policy uncertainty is larger than the prior month. There is evidence that the economic policy uncertainty influences return reversals through the liquidity channel. Short-term reversal profits are also positively related to the VIX index, the Baker-Wurgler (2007) investor sentiment index, and the Aruoba-Diebold-Scotti (2009) business conditions index in the prior month, though the predictability of the latter two indexes is less robust. However, adding these indexes and other variables does not weaken the relationship between economic policy uncertainty and return reversals.

Keywords
Asset pricing, economic policy uncertainty, return reversals

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Chapter 29: Time aggregation and the estimation of the market model: Revision and extension

Data for heavily and lightly traded firms are used to evaluate the effects of temporal aggregation on beta estimates, t values, and R^2 estimates. In addition to our analysis of the standard market model, dynamic market and error correction models are estimated. This study evaluates differences in the short-term and long-term dynamic relationships between the market and each type of firm. It is found that temporal aggregation has important effects on both the specification of a market model and the stability of beta estimates.

Keywords
Temporal aggregation, beta coefficients, dynamic market model, error correction model, coefficient determination, t statistics, long-run multiplier

Chapter 30: Leases on balance sheets

Leasing transfers effective ownership of physical assets to specialists. Firms not specializing in asset ownership focus on core competence and scale. In asset ownership including real estate, specialists have low or zero corporate tax rates and use depreciation to shield dividends. With operating leases, non-specialist firms operate the assets with no entry on balance sheets. On income statements, non-specialists deduct rent as an expense. On specialist income statements, the rent is exempt from double taxation of distributed dividends. Accounting reform proposes to undo and upend these arbitrage provisions. All leases have capitalized present values and are reflected as right-to-use assets and liabilities on balance sheets, expensed straight line. Reported liabilities rise, potentially undermining debt covenants. Sale and leaseback provisions are at risk of being reversed, notably if proceeds have been already distributed.

Keywords
Operating, capital and finance leases, on-balance sheet leases, off-balance sheet leases

Chapter 31: Financial econometrics, mathematics, statistics, and financial technology: An overall view

Based upon my experience in research, teaching, writing textbooks, and editing handbooks and journals, this review paper discusses how financial econometrics, mathematics, statistics, and financial technology can be used

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in research and teaching for students majoring in quantitative finance. A major portion of this paper discusses the essential content of Lee and Lee (2020). Then, Lee *et al.* (2019), Lee *et al.* (2017), and Lee and Lee (2015) are used to enhance the content of this paper. In addition, important and relevant papers, which have been published in different journals, are also used to support the issues discussed in this paper. I have found that the applications of financial econometrics, mathematics, statistics, and technology have improved drastically over the last five decades. Therefore, both practitioners and academicians need to update their skills in this area to compete in both financial market and academic research.

Keywords
Financial econometrics, financial mathematics, financial statistics, financial technology, machine learning, skewness, kurtosis

Chapter 32: Entropic two-asset option

This article extends the Margrabe formula such that it is suitable for accounting for *any* type jump of stocks. Despite the fact that prices of an exchange option are characterized by jumps, it seems that no study has explored those price jumps of an exchange option. The jump in this article is illustrated by a Poisson process. Moreover, the Poisson process can be extended into the Cox process in case there is more than one jump. The results illustrate that incompleteness in an exchange option leads to a premium which in turn increases an option value whilst hedging strategies reveal the mixed-bag type of results.

Keywords
B-S model, Cox process, exchange option, inverse Fourier transform, Poisson process

Chapter 33: Joint normality test for the returns on the futures and spot

It is well known that the optimal hedge ratios derived based on the mean–variance approach, the expected utility-maximizing approach, the mean-extended Gini approach, and the generalized semi-variance approach will all converge to the minimum variance hedge ratio if the futures price follows a pure martingale process and if the spot and futures returns are jointly normal. In this chapter, we perform empirical tests to see if the pure martingale and joint normality hypotheses hold using 25 different futures contracts

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and five different hedging horizons. Our results indicate that the pure martingale hypothesis holds for all commodities and all hedging horizons except for three stock index futures contracts. As for joint normality, we propose two new tests based on the generalized method of moments, which allow for calculating multivariate test statistics that take account of the contemporaneous correlation across spot and futures returns. Our findings show that the joint normality hypothesis generally does not hold except for a few contracts and relatively long hedging horizons.

Keywords
Hedge ratio, pure martingale hypothesis, joint normality hypothesis, pure martingale and joint normality hypotheses

Chapter 34: Analysis of theoretical and empirical relationships between the treasury bills and Eurodollar

In this paper, we derive an *equilibrium* relationship between the yields on Eurodollar and Treasury bills based on *equivalent martingale* results derived by Harrison and Kreps (1979) and Harrison and Pliska (1981, 1983), and *corporate debt pricing* model developed by Merton (1974). The derived *equilibrium* relationship incorporates the models used by Booth and Tse (1995) and Shrestha and Welch (2001) as special cases. The *equilibrium* relationship indicates that the conditional volatility of yield on Eurodollar explains the variation in TED spread. We empirically test the equilibrium relationship using a GARCH-M model and the concept of fractional cointegration. We use both the *ex ante* data implied by the respective futures contracts as well as the *ex post* spot data with daily, weekly, and monthly frequencies. We find empirical support for the *equilibrium* relationship.

Keywords
Stationary, fractional cointegration, TED spread

Chapter 35: Volatility risk measures and banks' leverage

In this paper, we investigate how different measures of volatility influence bank's capital structure besides mandatory capital requirements. We study the relationship between four volatility risk measures (volatility skew and spread, variance risk premia, and realized volatility) and bank's market leverage and we analyze if banks adjust their capital needs in response to a significant increase of risk premia discounting from traders. Among the four

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volatility measures, volatility skew (defined as the difference between OTM put and ATM call implied volatility and representing the perceived tail risk by traders) affects bank's leverage the most. As volatility skew increases — hence OTM put became more expensive than ATM call — banks deleverage their assets' structure. One plausible explanation relates to the higher costs of equity issuance that a bank will face during a period of distress. As the possibility to incur expensive equity issuance increases, the bank prefers to deleverage its balance sheet and create a capital buffer.

Keywords
Volatility risk measure, implied volatility, volatility skew, volatility spread, volatility risk premia, realized volatility, banks, capital requirements, options, derivatives, leverage

Chapter 36: The reactions to on-air stock reports: Prices, volume, and order submission behavior

The purpose of this paper is to analyze the real-time responses of stock prices, volume, and order submission behavior across investor groups to 2894 on-air stock reports from 9:16 a.m. to 1:15 p.m. during regular trading hours from 10/11/2010 to 12/31/2010 in Taiwan. First, positive (negative) reports move stock prices upwards (downwards) in real time, accompanied by increasing trading volume. However, the observed price movements are short lived and vanish 14 days afterwards. Second, responding to the reports, individual investors trade more actively and aggressively than institutional investors do. The overreaction of individual investors is responsible for the observed price movements.

Keywords
On-air stock reports, real-time responses, prices, volume, order submission behavior

Chapter 37: Mutual fund competition for ranking: When risk-taking comes with managerial effort

This study investigates theoretically and empirically mutual fund managers' risk-taking behavior due to ranking objectives. We argue that managers can not only choose the riskiness of their portfolio but also determine how hard to work (their effort). The combination of risk and effort depends on the interim performance gap and the effort cost level. Both interim winner and loser gamble by taking high risk and spending low effort when the interim performance gap is below a certain threshold. Only the interim loser gambles

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when the interim performance gap is small and the effort cost is sufficiently high. Otherwise, managers adopt the same choice of risk effort. In many cases, high (low) risk-taking induces higher (lower) effort. Empirically, we find that managerial effort is strongly and positively linked to their risk-shifting level. The worst performers behave differently from the others but are not necessarily riskier and lazier.

Keywords
Mutual fund, fund competition, excessive risk-taking, tournament

Chapter 38: Hedge ratios: Theory and applications

This paper first presents a review of various theoretical models and six estimation methods for the optimal futures hedge ratios. Then we use data to show how some of the hedge ratios can be applied to estimate hedge ratios in terms of S&P 500 future. We also show the estimation procedure on how to apply OLS, GARCH, and CECM models to estimate optimal hedge ratios through R language. These approaches are theoretically derived in terms of minimum variance, mean–variance, expected utility, and value-at-risk. Various ways of estimating these hedge ratios are also discussed, ranging from simple ordinary least squares to complicated heteroscedastic cointegration methods. Under martingale, joint normality conditions, and some other conditions, different hedge ratios can be shown that this different ratio can be converted to the minimum variance hedge ratio. Otherwise, the optimal hedge ratios based on the different approaches are in general different. Finally, our empirical findings suggest the importance of capturing the heteroscedastic error structures and including the long-run equilibrium error term in the conventional regression model.

Keywords
Hedge ratio, minimum variance hedge ratio, optimum mean–variance hedge ratio, sharpe hedge ratio, GARCH method, R language

Chapter 39: A note on stock market seasonality: The impact of stock price volatility on the application of dummy variable regression model

This article provides both statistical analysis and empirical evidence that the dummy variable regression models extensively employed in the market

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seasonality literature may wind up misleading results. We show that the estimates of the said model tend to reject the null hypothesis incorrectly once the stock returns exhibit higher volatility for the specified period under examination. Our empirical results suggest that the so-called "January effect" could be attributed to the application of the inappropriate statistical method.

Keywords
Stock market seasonality, dummy variable regression model, January effect

Chapter 40: Time-changed GARCH versus GARJI model for extreme events: An empirical study

In the literature, a GARCH-jump mixture model, namely, the GARCH-jump model with autoregressive conditional jump intensity (GARJI) model, in which two conditional independent processes, i.e., a diffusion and a compounded Poisson process, are used to account for stock price movements caused by normal and extreme event news arrivals, individually, is developed by Chan and Maheu (2002, 2004) to describe the volatility clustering and leverage effect phenomenon. The resulting model is less efficient and provides only *ex post* filter for the probability of the occurrences of large price movements. A more informative and parsimonious model, however, the VG NGARCH model, is proposed and calibrated in this study. Being an extension of the variance gamma model developed by Madan *et al.* (1998), the proposed VG NGARCH model incorporates an autoregressive structure on the conditional shape parameters, which describes the news arrival rates of different impact sizes on the price movements, and an *ex ante* prediction for the occurrences of large price movements is provided. The performance of the proposed VG NGARCH model is compared to the GARJI model based on daily stock prices of five component financial companies in S&P 500, namely, Bank of America, Wells Fargo, J.P. Morgan Chase, CitiGroup, and AIG, respectively, from January 3, 2006, to December 31, 2009. The goodness of fit of the VG NGARCH model and its ability to predict the probabilities of large price movements are demonstrated by comparing with the benchmark GARJI model.

Keywords
GARCH-jump mixture model, GARJI model, diffusion process, compounded Poisson process, volatility clustering, leverage effect, *ex post* filter, VG NGARCH model, variance gamma model, shape parameter, *ex ante* prediction, goodness of fit

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Chapter 41: Corporate financial hedging and the cost of equity capital

Using a large panel of UK public firms, we examine the relationship between the financial risk hedging and the cost of equity capital. We hypothesize that firms utilizing financial derivative instruments reduce the stock return volatility which is priced in investors' expectations. While financial risk hedging serves as a vehicle for firms to alleviate cash flows volatility, it also leads to economic benefits to the firm value in case of the presence of increasing asymmetric information. In addition, we hypothesize and test whether the nature of relation between financial risk hedging and cost of equity capital varies and is more negative or more ambiguous with economic shocks. Our results show that engaging in financial risk hedging enables firms to have a lower cost of capital. Consistent with the extant literature, we control for potential endogeneity problems and sample selection bias using instrumental variables and treatment effects approaches. Thus, our results are robust to a battery of sensitivity checks, including the use of multiple estimation methods and alternative proxies of cost of equity measures. Overall, our findings suggest that the value of financial hedging decisions increases during economic shocks, if financial constraints become more severe, and if cash flows volatility increases.

Keywords
Financial risk hedging, stock volatility, information asymmetry, cost of equity capital, UK firms

Chapter 42: Does trading volume contain information to predict stock returns? Evidence from China's stock markets

This paper examines empirical contemporaneous and causal relationships between trading volume, stock returns, and return volatility in China's four stock exchanges and across these markets. We find that trading volume does not Granger-cause stock market returns on each of the markets. As for the cross-market causal relationship in China's stock markets, there is evidence of a feedback relationship in returns between Shanghai A and Shenzhen B stocks, and between Shanghai B and Shenzhen B stocks. Shanghai B return helps predict the return of Shenzhen A stocks. Shanghai A volume Granger-causes return of Shenzhen B. Shenzhen B volume helps predict the return of Shanghai B stocks. This paper also investigates the causal relationship among these three variables between China's stock markets and the U.S.

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stock market and between China and Hong Kong. We find that U.S. return helps predict returns of Shanghai A and Shanghai B stocks. U.S. and Hong Kong volumes do not Granger-cause either return or volatility in China's stock markets. In short, information contained in returns, volatility, and volume from financial markets in the U.S. and Hong Kong has very weak predictive power for Chinese financial market variables.

Keywords
Spillover, China's stock markets, trading volume

Chapter 43: Financial statement analysis

This chapter provides readers with the skills and insights to enable a fuller understanding of a public company's underlying financial statements. Financial statement analysis is a form of fundamental analysis that identifies and analyzes the key financial information relevant to a company, for the purpose of determining the company's intrinsic value. This chapter concentrates on the calculation and application of the principal financial ratios for profitability, efficiency, liquidity, solvency, and investment potential. There is an effort to make the content accessible to readers on a practical level, by applying the various ratios to the real financial statements of Johnson & Johnson. While the chapter outlines in detail the quantitative calculations required to undertake financial statement analysis, there is a firm emphasis on the importance of context in properly applying these techniques. The establishment of trends, the comparison of data to benchmarks, and the consideration of strategic factors all form part of this holistic contextual analysis of financial statement information.

Keywords
Ratio analysis, profitability, efficiency, liquidity, solvency, investment potential

Chapter 44: Expected credit losses under IFRS 9: Concept, models, and disclosures

The IFRS 9 on financial instruments has made an important contribution to the credit loss recognition process and financial reporting by replacing the existing Incurred Credit Loss (ICL) model with the Expected Credit Losses (ECL) model. The ECL model applies to all financial instruments whether they are recognized at the amortized cost or at fair value. Firms are required to estimate and recognize loan loss allowances based either on

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the 12-month or lifetime expected credit losses, depending on whether there has been a significant increase in the credit risk since initial recognition. In this chapter, we first briefly explain the scope of IFRS 9 and then discuss the main characteristics of ECL model and also present mathematical models that can be used to estimate credit loan losses. The mathematical models can be based either on the capital market, discounted cash flow, or weighted losses approach. Finally, we discuss ECL disclosures that are expected to provide greater transparency on credit risk and loan loss provisions, and also present economic implications of the ECL model on firm performance.

Keywords
IFRS 9, IAS 39, Expected Credit Loss (ECL), Incurred Credit Loss (ICL), Basel Committee, credit risk, default, provision for loan losses

Chapter 45: Hedging with the international equity index futures: The conventional model versus the error correction model

This study estimates and compares the hedge ratios of the conventional and the error correction models using three advanced international stock markets with different time intervals. Comparisons of out-of-sample hedging performance reveal that the error correction model outperforms the conventional model, suggesting that the hedge ratios obtained by using the error correction model do a better job in reducing the risk of the cash position than those from the conventional model. In addition, this paper evaluates the effects of temporal aggregation on hedge ratios. It is found that temporal aggregation has important effects on the hedge ratio estimates.

Keywords
Hedge ratio, conventional approach, error correction approach, hedging performance, temporal aggregation

Chapter 46: Technical analysis in investing

Technical analysis helps investors to better time their entry and exit from financial asset positions. This methodology relies solely on past information of financial assets price and volumes to predict a financial asset's future price trend. Modern research has established that combined with other sentiment measures, such as social media, it can outperform the standard buy and hold

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strategy. Moreover, it has been documented that novice and professional investors have technical analysis in their investing strategy. An experienced investor should combine fundamental analysis and technical analysis for better trading results. Programmers use technical analysis to create algorithmic trading systems that learn and adopt to the changing trading environments and perform trading accordingly without human involvement. There are hundreds of technical tools offered by known trading platforms. Investors must use specific tools that fit their trading style and risk adoption. Moreover, different financial assets, such as stocks, ETFs, cryptocurrency, futures, and commodities, demand different sets of tools. Furthermore, investors should use these tools according to the time frame they use for trading. This chapter will discuss different technical tools that are used to help traders of different time frames and different financial assets to achieve better returns over the traditional buy and hold strategy.

Keywords
Technical analysis, oscillators, support and resistance, Bollinger Bands, linear regression, trend lines

Chapter 47: Venture capital investment valuation — Skewness of returns

Based upon the comparative analysis, we first discuss different kinds of Greek letters in terms of the Black–Scholes option pricing model and then we show how these Greek letters can be applied to perform hedging and risk management. The relationship between delta, theta, and gamma is also explored in detail.

Keywords
Delta (Δ), theta (Θ), gamma (Γ), vega (ν), rho (ρ), hedging

Chapter 48: A correlation-based portfolio choice algorithm

Analyzing the correlation matrix of listed stocks, we identify "singletons" that table minimal cross-sectional correlations. Portfolios comprised of 100 to 500 singletons all have lower betas and standard deviations and, correspondingly, higher average Sharpe and Treynor ratios than the CRSP (Center for Research in Security Prices) universe over the sample time period 1950–2017. Portfolios of singletons chosen from subsets of the CRSP universe, including

small-value, low-variability, and momentum stocks, similarly realize lower portfolio standard deviations and higher risk-adjusted returns. These well-diversified portfolios suggest that the positive abnormal returns to low-beta portfolios are driven by their component stocks having low average cross-sectional correlation. One of the authors invested $20,000 of their own money in the algorithm-chosen 240 stock singleton portfolio over a four-year period (2015–2018) and beat the market year by year on a risk-adjusted basis just as our results predicted.

Keywords
Low cross-sectional correlation, portfolio choice, diversification, return comovement, low-variability anomaly, betting against beta

Chapter 49: Stock returns and volatility on China's stock markets

We examine time-series features of stock returns and volatility, as well as the relation between return and volatility in four of China's stock exchanges. Variance ratio tests reject the hypothesis that stock return follows a random walk. We find evidence of long memory of returns. The application of GARCH and EGARCH models provides strong evidence of time-varying volatility and shows that volatility is highly persistent and predictable. The results of GARCH-M do not show any relation between expected returns and expected risk. Daily trading volume used as a proxy for information arrival time has no significant explanatory power for the conditional volatility of daily returns.

Keywords
Volatility, China's stock markets, EGARCH

Chapter 50: Value line investment survey rank changes and beta coefficients

By using a regression relationship, this chapter investigates the relationship between value line rank changes and beta changes in an attempt to explain the value line enigma.

Keywords
Beta coefficient, value line ranking, value line enigma, market model

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Chapter 51: International hedge ratios for index futures market: A simultaneous equations approach

The main purpose of this chapter is to investigate hedge ratios in terms of the international index futures markets. Instead of looking at hedging in a single market, we here construct a simultaneous equations system to study the index hedging in the light of the cross-country linkage and interaction. The three-stage least squares (3SLS) estimating procedure is then applied to S&P 500, FTSE100, and NIKKEI225 indices over the period 1990–2020. The empirical results indicate that the cross-country hedging strategy in both markets is feasible and the investors can bring down the holding position in their own futures market. Moreover, the hedging effectiveness of cross-country hedging strategy performs better than the traditional single market hedging strategy in terms of the percentage reduction in variance.

Keywords
Hedge ratio, cross-country linkage, simultaneous equation, three-stage least squares

Chapter 52: Empirical studies of structural credit risk models and the application in default prediction: Review and new evidence

This paper first reviews empirical evidence and estimation methods of structural credit risk models. Next, an empirical investigation of the performance of default prediction under the down-and-out barrier option framework is provided. In the literature review, a brief overview of the structural credit risk models is provided. Empirical investigations in extant literature papers are described in some detail, and their results are summarized in terms of subject and estimation method adopted in each paper. Current estimation methods and their drawbacks are discussed in detail. In our empirical investigation, we adopt the Maximum Likelihood Estimation method proposed by Duan (1994). This method has been shown by Ericsson and Reneby (2005) through simulation experiments to be superior to the volatility restriction approach commonly adopted in the literature. Our empirical results surprisingly show that the simple Merton model outperforms the Brockman and Turtle (2003) model in default prediction. The inferior performance of the Brockman and Turtle model may be the result of its unreasonable assumption of the flat barrier.

Keywords
Structural credit risk model, estimation approach, default prediction, maximum likelihood estimation (MLE)

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Chapter 53: Predicting stock return movement directions with sentiment analysis of news headlines: A machine learning approach

This paper combines the sentiment features of news headlines, stock market data, and macroeconomic factors to predict the direction of stock return movements one month after the release of the news article. The sentiment features are extracted with Flair, advanced libraries for Natural Language Processing (NLP). The stock market data of a company contain a comprehensive collection of 94 variables used by *a priori* literature (Gu, Kelly, and Xiu, 2020) for the prediction of stock return. To construct the prediction model, this paper applies seven machine learning algorithms (including gradient boosting, XGBoosting, random forest, artificial neural networks, support vector machine, naïve Bayes, and logistic regression). The out-of-sample tests show that tree-based ensemble methods (i.e., random forest, XGBoosting, and gradient boosting) provide the most accurate predictions, with a maximum AUC-ROC of 0.74. Furthermore, this study provides evidence for the effectiveness of the sentiment features of news headlines for the prediction of future stock returns as the median of the sentiment score of the news headline is listed as one of the most important predictors in the model.

Keywords
Stock return directions, machine learning, sentiment analysis, textual analysis, natural language processing, news headlines

Chapter 54: Style investing, momentum, and comovement

Investors and professional money managers typically categorize assets into different styles to facilitate portfolio management and capital allocations. As these market participants move funds among assets of different styles based on their relative performance, correlated trading generates return comovement and style momentum. This chapter reviews the existing theories on style investing and important findings. In particular, it presents new evidence in a large bond market and demonstrates that behavioral finance theory can

help explain return comovement and momentum in the bond market traditionally dominated by institutional and long-term investors who are thought to be less behaviorally biased.

Keywords
Style investing; comovement, style momentum, behavioral biases, rating recalibration, return predictability, systematic risk, asset pricing

Chapter 55: Mining for "green diamonds" — Value relevance of greenhouse gas emissions

Using an international dataset of 5,861 firm-year observations between 2009 and 2016 obtained from the Carbon Disclosure Project (CDP), we analyze the effect of firms' Greenhouse Gas (GHG) emissions on stock price performance. To this end, we first discuss former research which finds an equity discount entailed by high levels of GHG emissions. We then focus on additional metrics of stock price performance, namely, stock price return and stock price risk. Interestingly, we do not find any significant impact of GHG emissions on these metrics. A possible explanation is that investors are not yet able to quantify the GHG emission risk due to insufficient disclosure.

Keywords
Greenhouse gas, carbon emissions, climate change, firm value, stock return, market risk, value relevance, Carhart four-factor model

Chapter 56: Risk estimation, diversification, and optimal weights

The main purposes of this chapter are (i) to discuss risk classification and estimation, (ii) to show how to use minimum variance and Sharpe performance measure approach to estimate optimal weights for a two-security portfolio, (iii) the applications of performance measures that are also discussed, and (iv) to use concepts discussed in this chapter to show how banking lending rate can be estimated.

Keywords
Business risk, financial risk, diversification, beta coefficient, Sharpe performance measure, optimal weights, banking lending rate

Chapter 57: The role of founder presence in investment analysis

Extant studies and my own work show that in U.S. listed corporations, the presence of a firm's founder adds value to the firm. The incremental value

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increases with the extent of decision rights controlled by the founder. Furthermore, the value addition is higher if the founder CEO is younger at the time of the initial public offering and decreases with the founder's tenure in the firm. Further investigation reveals that the founders add value by improving operating performance and being more transparent than similar non-founder firms. Moreover, the founders are more focused on the strategic positioning of the firm in improving operating performance — they improve profit margins in differentiated firms while improving efficiency in firms with a cost leadership strategy. Analysts and investors can benefit by incorporating these insights into their analysis.

Keywords
Founder firms, family firms, founder influence, founder tenure, founder horizon, operating performance, transparency, strategy

Chapter 58: Financial statement analyses and firm valuation: Johnson and Johnson (JNJ) as a case study

The main purposes of this chapter are to use Johnson and Johnson's accounting information and market information to discuss the following three issues for investment analysis. These three issues are (i) financial ratio analyses, (ii) impact of intangible assets on Tobin Q estimates, and (iii) to show how a simultaneous model can be used to forecast pro forma financial statements. Peter and Taylor (2017) have theoretically shown that intangible asset is one of the most important in calculating Tobin Q. We use accounting information to calculate Tobin Q with and without considering intangible assets. Since JNJ's intangible assets count about 51%, we review the Tobin Q's alternative estimation and show how the intangible asset affects Tobin Q in JNJ. By using financial ratios and market information, we perform 20 equation models to forecast the pro forma balance sheet and pro forma income statement.

Keywords
Liquidity ratios, financial leverage ratios, asset efficiency ratios, profitability ratios, market value ratios, policy ratios, financial z-score, Tobin Q

Chapter 59: Technical analysis in the stock market: A review

Technical analysis is the study of forecasting future asset prices with past data. In this survey, we review and extend studies on not only the time-series

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predictive power of technical indicators on the aggregated stock market and various portfolios but also the cross-sectional predictability with various firm characteristics. While we focus on reviewing major academic research on using traditional technical indicators, we also discuss briefly recent studies that apply machine learning approaches, such as Lasso, neural network, and genetic programming, to forecast returns both in the time series and on the cross-section.

Keywords
Technical analysis, machine learning, genetic programming, cross-sectional returns, predictability

Chapter 60: The sovereign rating channel in the European debt crisis: Spillover effects on sovereign CDS and other systemic risk indicators

This paper investigates the effect of sovereign debt ratings on Credit Default Swap (CDS) spreads during the Eurozone sovereign debt crisis. The empirical investigation is conducted by means of panel vector autoregressive models which allow the analysis of multidirectional relationships in a dynamic context. Our main findings, based on directional spillover effects, are that credit rating announcements did not have any impact on the CDS spreads of the Eurozone periphery countries. However, when we limit our analysis of the sovereign debt ratings of Portugal and Ireland to a minor extent, we obtain some evidence for spillover effects on the CDS spreads of other countries. Furthermore, the quantitative contribution of sovereign ratings has been of secondary importance when calculating their spillover impact on other "systemic" risk indicators. Overall, the results indicate that during the recent sovereign debt crisis, credit rating announcements on the Eurozone's periphery countries did not contribute to the global financial crisis.

Keywords
Sovereign risk ratings, volatility spillover effects, European sovereign debt crisis

Chapter 61: Interest rate sensitivity and investor disagreement: How to explain bank stock turnover

Maturity mismatches (MMs) expose banks to interest rate sensitivity, adding to the uncertainty of banks' performances. Since information regarding maturity mismatches is usually not readily available, considering the high correlation between the two, interest rate sensitivities could serve as proxies

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for these mismatches for short periods. Therefore, we label them as implied maturity mismatches. Our analyses of the correlation between banks' interest rate sensitivity and the trading volume of the banks' equity reveal positive correlations. The heightened increase in volume suggests that implied maturity mismatches increase disagreement among banks' investors.

Keywords
Interest rate sensitivity, maturity mismatches, trading volume, investors' disagreement, analysts' forecast dispersion

Chapter 62: A novel semi-static method for the index tracking problem

With the rapid growth of index investing in wealth management, such as the exchange traded fund (ETF) market, tracking the reference index of ETF regains considerable attention because the corresponding tracking error is one of the key measurements of a fund's performance. To improve accuracy and efficiency in minimizing tracking errors, we propose a novel approach that combines the generalized singular value decomposition (GSVD) and the constrained Kalman filter method for estimation of portfolio weights and their dynamical updates, respectively. GSVD reveals all feasible weights under a static constrained regression model for the index tracking error minimization problem. The constrained Kalman filter is applied to update the GSVD-solved weights of the optimal portfolio along with the dynamical time series of return data. This two-stage approach is semi-static. Our empirical studies demonstrate that sufficient low tracking error can be obtained under a robust top–down stratified framework that consists of eleven sector indices defined by GICS.

Keywords
Generalized singular value decomposition (GSVD), constrained Kalman filter method, robust index tracking

Chapter 63: Fundamental analysis: A practical approach

Fundamental analysis is a simple concept which tries to isolate investing choices from sentiment. Moreover, it is a mixture of science and art that breaks down to simple math. Great value investors like Warren Buffet, Benjamin Graham, and John C. Bogle very vividly state that in order to successfully value a business, it does not take more than common sense and simple math. In this chapter, we approach the very fundamentals in a

practical manner, showcasing a simple way to estimate the intrinsic value of a business.

Keywords
Fundamental analysis, value investing, discounted cash flows (DCF), intrinsic value

Chapter 64: Lessons on risk, return, and portfolio construction from the great investors

There is often a wide divergence between academic and practitioner views on risk, return, and portfolio construction. For example, academics focus primarily on purely quantitative measures or factors. Initially, the focus was on dividends, free cash flow, standard deviation, and beta. Later, additional factors analyzed by the academic community came into focus, such as size, style, liquidity, momentum, and quality. Practitioners, in contrast, often focus on a company's products, its history, and the competitive dynamics of its industry. Furthermore, practitioners "discovered" anomalies, such as momentum, decades before they were rigorously analyzed and published by academics. The current distinction between the two groups is not merely quantitative versus qualitative. This chapter will summarize the viewpoints of the two camps — academic and practitioner — and suggest steps that may effectively combine the two schools of thought, at least to a certain degree using the Black–Litterman model and other qualitative techniques, such as stratifying asset pricing models. This analysis may result in a more robust investment, risk management, and portfolio construction process.

Keywords
Great investors, investment strategies, modern portfolio theory, factor models, Warren Buffett, Benjamin Graham, James Simons, Cathie Wood, Black–Litterman model

Chapter 65: Sources of the liquidity premium: Risk or mispricing?

We study three widely used liquidity measures and find that they all carry significant premiums beyond the size, book-to-market, and momentum effects. Although liquidity as a risk factor bears a significant return premium, it is better characterized by a characteristic-based model. Further analysis shows that (1) although the premium persists for up to five years following formation, it diminishes over time and becomes insignificant in the post-1960 period and (2) the premium is larger for stocks with higher idiosyncratic risk. Thus, the empirical results provide some evidence that supports the mispricing argument.

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Keywords
Liquidity, factor model, characteristic-based model, risk, mispricing

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Chapter 66: Analysis of IBEX-35 listed companies: Recent CSR reports and behavior of the main indicators. Existence of a proportional relationship between greenwashing and deficient CSR reports

Currently, environmental concerns, as well as other issues such as Corporate Social Responsibility (RSC) or socially responsible investment (SRI), are leading many companies to implement "greenwashing" behaviors. Throughout this article, the factors of "greenwashing" will be analyzed, and the relationship between those behaviors and deficient sustainability reports will be examined. This study will be approached from the point of view of the Spanish reality, paying special attention to listed companies, specifically those which are listed on the IBEX 35 Index. In order to achieve that objective, this article will analyze the sustainability reports of six companies listed on the IBEX 35 Index belonging to different sectors: energy, oil, consumer goods, financial services, basic materials, industry and construction, and technology and telecommunications. Moreover, greenwashing in the plastic industry will be analyzed. Finally, it will be possible to draw some conclusions about the relationship between "greenwashing" and CSR.

Keywords
Circular economy, greenwashing, socially responsible investment (SRI), Corporate Social Responsibility (RSC), CSR reports, sustainability reports, IBEX 35 Index, non-financial information, plastic industry

Chapter 67: Return volatility, skewness, and momentum effects

This paper identifies non-linear patterns of momentum profits across stocks with different levels of return volatility and skewness. It finds that momentum profits are the largest from mildly volatile and skewed stocks; this phenomenon is consistently observed for different formation/holding periods, types of examined returns, and momentum grouping. According to the patterns, this paper proposes a sample filtering criterion for momentum investment; the profits accrued through the sample filtering are economically enlarged. Such enhancement of momentum profits by the filtering process is documented in various strategies, including the conventional, 52-week high, and risk-managed momentum strategies.

Keywords
Momentum, volatility, skewness, sample filtering

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Chapter 68: Predicting implied volatility with historical volatility

In this study, we document a novel lead–lag relation between historical and implied volatilities based on China's CSI300 Index options. We show that historical volatilities have incremental information when we predict implied volatilities, and this pattern tends to be more stable for put options than for call options. Moreover, we reveal that this lead–lag relation is relevant to option terms and time horizons of historical volatilities, which means implied volatilities of long-term options are more like to be properly predicted by long-term historical volatilities on average. Finally, we find that speculative trading might explain our results.

Keywords
Option, implied volatility, historical volatility, lead–lag relation

Chapter 69: Estimating binomial and Black and Scholes option pricing models: Excel, R language, and SAS program approach

In this chapter, we showed how Excel and R language programming can be used to estimate the European call/put prices based on the Black–Scholes model as well as binomial option pricing model. Different underlings are considered, including individual stocks, currency, and stock indices. SAS language programming to estimate the European call/put prices based on the binomial option pricing model and Black–Scholes option pricing model is given in the appendices.

Keywords
Excel, R language programming, European call/put, Black–Scholes model, binomial option pricing model, stock indices, SAS language programming

Chapter 70: Value contributions

When firms increase the size of their net balance sheet, they earn a higher return as value. Value is earned from different holdings. An examination of holdings compares which earn value. Holdings of US-traded firms are divided into categories from their balance sheet. As reported on the asset side are cash, receivables, physicals, and intangibles. On the liability side in order

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are payables, short- and long-term debt. In value, the sum of book equity is divided by market capitalization. For individual value, each is divided by market capitalization. Then, the return is regressed on the individual values and their interactions. Negative value is when increasing a holding reduces the return. Positive value comes from then return increasing with a holding. The rank order of contributors to value, for US firms from 1980–2016, is

(Short-Term Debt)
Receivables
Cash
(Payables)
(Long-Term Debt)
Physicals
Intangibles.

Replacing holdings by their risk-weight adjustments and adding up, the resulting book-to-market has no effect on returns. Returns are fully priced by firms' balance sheet holdings. The results potentially explain why value or overall book-to-market fails to outperform growth. Value is a return to holding certain risky holdings. Not all balance sheet holdings are risky.

Keywords
Financial accounting, asset pricing, balance sheet, cash, short-term debt, trade credit

Chapter 71: Using computational science methods in accounting and finance research

Computational modeling is being used in more research and applications in accounting and finance. These approaches include simulations and computer intensive methods that are different than traditional methods in most prior research. This chapter explains the underlying philosophy of these methods and discusses several research studies using them. Inferences of the studies are discussed and both benefits and concerns with the methods are addressed.

Keywords
Computational modeling, computational science, simulation, computer intensive methods, Monte Carlo simulation

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Chapter 72: Stock buybacks and financial turmoil: Pros and cons for investors

Buyback programs are often used by firms for different purposes, including distributing excess cash to shareholders and signal that the stock price is underpriced. The first purpose of this chapter book is to review studies of buyback programs and to highlight that fundamentals-based hypotheses are problematic in financial turmoil. We will show how buyback programs add value to shareholders while also identifying some situations in which they can destroy value. The second purpose is to present the pros and cons of buyback programs to shareholders, particularly during financial turmoil.

Keywords
Payout policy, stock repurchase, financial crisis, insider trading, financial flexibility

Chapter 73: The Roles of Financial Analysts in the Stock Market

Investors lacking ample time, professional knowledge, and sufficient ability may find it difficult to understand the implication of complex corporate information and figure out the clear trend of future corporate performance. Financial analysts who provide earnings forecasts and stock recommendations could help investors with investment decision-making. This chapter explores the roles that analysts play in the stock market, the determinants of the effectiveness of their roles, and how well they play the roles.

Keywords
Financial analysts, analyst coverage, analyst forecasts, information intermediation, external monitoring

Chapter 74: Funding liquidity and CDS-bond basis: Evidence from the CDS Big Bang

The CDS Big Bang increased the upfront funding requirements for trading CDS contracts, especially for those with credit spreads further away from 100 and 500 basis points. Exploiting this regulatory change, we document that a higher funding requirement reduces market liquidity and increases the absolute value of the CDS-bond basis. The funding effects are stronger

for smaller reference entities. Our findings highlight the importance of the trade-off between standardization and the funding cost of upfront payments.

Keywords
CDS-bond basis, CDS Big Bang, funding liquidity

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Chapter 75: Issues and challenges of weather and freight derivatives: Impact of pandemic situation

During the last couple of months, the pandemic situation raised the need for assessment of the impact on derivatives particularly weather and freight derivatives, as an innovative financial product. There are several issues and challenges faced by weather and freight derivatives in the financial market. This paper aims to appreciate innovative financial derivatives and also addresses issues relating to the functioning of weather and freight derivatives. We have also examined the pricing models of weather derivatives across the globe.

Objectives:

1. To understand the pricing models of weather and freight derivatives in global context.
2. To examine the impact of the COVID-19 pandemic situation on weather derivatives.

Keywords
Financial markets, financial derivatives, weather derivatives, freight derivative, hedging

Chapter 76: On a long-term investment strategy in a stock market

Against the backdrop of increasingly fierce industrial competition nowadays, firms tend to have substantive business risk and/or information risk, increasing the estimation risk and limit of arbitrage for investors in their short-term investments in a stock market. It is thus important for investors to hold a long-term horizon for at least part of their investments in the stock market. This chapter aims to introduce a long-term investment strategy that is practically feasible and potentially profitable for investors. To this end, we first develop a parsimonious model in which we identify the major determinants of a firm's value and long-term growth. This model is used to select

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high-value firms from each industry for further fundamental analysis and valuation. We next expatiate on how to perform strategy analysis, accounting analysis, financial analysis, and prospective analysis, and therein apply the residual operating income valuation model, in the best possible manner to further value the selected firms and their long-term investment potential. Lastly, we expound on the strategy of forming and adjusting a long-term investment portfolio in a way that potentially maximizes long-term portfolio return.

Keywords
Long-term investments, determinants of firm value, strategy analysis, accounting analysis, financial analysis, prospective analysis, residual operating income valuation model, investment portfolio formation, adjustments

Chapter 77: European option, American option, and option bounds: Theory, method, and some empirical results

It is well known that both normal and log-normal distributions are important to understand Black & Scholes type European and American options. Therefore, we first review the basic theory of normal and log-normal distribution and their relationship, and then bivariate and multivariate normal density functions are analyzed in detail. Next, we discuss American options in terms of random dividend payment. We then use bivariate normal density function to analyze American options with random dividend payment. Excel programs are used to show how American co-options can be evaluated. Finally, pricing option bounds are analyzed in some detail.

Keywords
Normal distribution, log-normal distribution, American option, option bound, multivariate normal distribution, multivariate log-normal distribution

Chapter 78: Improving the stock market prediction with social media via broad learning

This chapter discusses how to exploit various Web information to improve the stock market prediction. We first discuss the impacts of investors' social network on the stock market and then propose several information fusion methods, that is, the tensor-based model and the multiple-instance learning model, to integrate the Web information and the quantitative information to improve the prediction capability.

Keywords
Stock prediction, event extraction, information fusion, social network, sentiment analysis, quantitative analysis, stock correlation

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Chapter 79: Bond portfolio management, swap strategy, duration, and convexity

This chapter first focuses on the bond strategies of riding the yield curve and structuring the maturity of the bond portfolio in order to generate additional returns. This is followed by a discussion of swapping, which are essentially interest rate swaps. Next is an analysis of duration or the measure of the portfolio sensitivity to changes in interest rates with and without convexity, after which immunization is the focus. The convexity is essentially discussed in the non-linear relationship between bond price and duration. Finally, a case study is presented on bond portfolio management in the context of portfolio theory. Overall, this chapter presents how interest rates change affects bond price and how maturity and duration can be used to manage portfolios.

Keywords
Bond strategies, swapping, substitution swap, intermarket spread swap, interest rate anticipation swap, pure yield pickup swap, duration, maturity

Chapter 80: Do CFA charterholders make better hedge fund managers?

In this paper, we investigate whether hedge fund managers possessing advanced professional education, as represented by the CFA@ designation, have better managerial ability and fund performance compared to managers lacking such education. We define managerial ability in terms of market, volatility, and liquidity timing ability. In general, we find that CFA-designated managers have better volatility and liquidity timing ability and earn better risk-adjusted returns than non-CFA managers. We also find that the CFA designation does not contribute to the fund flow-performance relation. The rise of high frequency and algorithmic trading may partially explain these findings.

Keywords
CFA, market timing, volatility timing, liquidity timing, hedge funds, fund flow, fund performance, fund management

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Chapter 81: The impact of bank activity and funding strategies on liquidity management: International evidence

This study examines the implications of a bank's activity mix and funding strategy for its liquidity management as defined in the Basel III accords or when they create more liquidity as measured by Berger and Bouwman (2009). Using an international sample of 624 banks in 65 countries, we find that, at low levels of non-interest income and non-deposit funding, there could be some risk diversification benefits of increasing these shares, though, at higher levels of non-interest income and non-deposit funding shares, further increases result in higher illiquidity. Finally, better accounting disclosure will improve the effect on bank liquid by both bank's activity mix and funding strategy, but at the same time, they worsen the bank liquid by stricter capital stringency and more market power.

Keywords
Bank, liquidity, non-interest income share, wholesale funding, global financial crisis

Chapter 82: Accounting information and firm valuation

Our article aims to elucidate the theoretical relation and empirical observation between accounting information and contemporaneous firm value (measured as a firm's stock price and/or stock return). Grounded on the intricate links of future expected cash flow and current stock price, future expected cash flow and future earnings, and future earnings and realized earnings, our work provides constitutional understanding of the residual income valuation model (Ohlson, 1995) and the abnormal earnings growth model (Ohlson and Juettner-Nauroth, 2003). The dynamics of the three intricate links and their implications for the valuation models have untangled a long-time myth in the literature of value relevance of financial reporting. Our article also presents the mediation effect of market efficiency on the empirical observation between accounting information and contemporaneous firm value.

Keywords
Accounting information, firm valuation, the residual income valuation model, the abnormal earnings growth model, value relevance, market efficiency

Chapter 83: Developments in CDS markets: A review on recent CDS studies

The Credit Default Swap (CDS) market has experienced tremendous changes over the last two decades. This paper reviews recent studies on the CDS

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contract and the CDS market, highlighting recent developments in CDS contracts and market structures. Characteristics of CDS contracts make them outstanding from other derivatives products, and the rapid development of CDS markets shows the popular use and importance of CDS contracts in general. Policies and regulations on CDS have brought huge changes to the market and the availability of publicly reported data. The relationship between firms and CDS has long been a focus among academics. Moreover, CDSs are also commonly used among mutual funds, hedge funds, and banks, affecting the trading activities of various financial intermediaries. While the sovereign CDS and index CDS are shown to be more and more significant in financial markets, other CDS products, including CDS options and tranches, also experience changes. We summarize the evolution in CDS markets and the structure of CDS studies, aiming to bring a better understanding of CDS markets and provide insights for future studies on the CDS market.

Keywords
Credit default swaps, OTC derivatives, credit risk, liquidity, asset pricing, market structure, sovereign debt

Chapter 84: Decision tree and Microsoft Excel approach for option pricing model

The main points of this chapter are (i) to use the decision tree approach to derive the binomial option pricing model (OPM) in terms of the method used by Rendleman and Barter (RB, 1979) and Cox *et al.* (CRR, 1979) and (ii) to use Microsoft Excel to show how the decision tree model can be converted to the Black–Scholes model when the number period increases to infinity. In addition, we develop a binomial tree model for American option and the trinomial tree model. The efficiency between binomial and trinomial trees is also compared. In sum, this chapter shows how binomial OPM can step by step be converted to Black–Scholes OPM.

Keywords
Binomial option pricing model, call option, put option, one-period OPM, two-period OPM, N-period OPM, synthetic option, Excel program, Black–Scholes model, Excel VBA, European options, American option

Chapter 85: Comparisons between the Markowitz model and the Black–Litterman model

As a sophisticated modification, the Black–Litterman portfolio model allows the optimal portfolio's weight to rely on a combination of the implied market

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equilibrium returns and investors' views (Black and Litterman, 1991). This chapter contrasts the Markowitz (1952) with the Black–Litterman model and includes the idea of Longo (2021). However, the performance of a Black–Litterman model is closely related to investors' views and the estimated covariance matrix. To overcome these problems, we first propose a predictive regression to form investors' views, where asset returns are regressed against their lagged values and the market return. Second, motivated by stylized features of volatility clustering, heavy-tailed distribution, and leverage effects, we estimate the covariance of asset returns via heteroscedastic models. Empirical analysis using five industry indexes in the Taiwan stock market shows that the proposed portfolio outperforms existing ones in terms of cumulative returns.

Keywords
Markowitz modern portfolio theory, Black–Litterman model, GARCH model, EGARCH model, the investor's views, volatility clustering

Chapter 86: Empirical performance of the constant elasticity variance option pricing model

In this essay, we empirically test the Constant Elasticity of Variance (CEV) option pricing model by Cox (1975, 1996) and Cox and Ross (1976), and compare the performances of the CEV and alternative option pricing models, mainly the stochastic volatility model, in terms of European option pricing and cost accuracy-based analysis of their numerical procedures.

In European style option pricing, we have tested the empirical pricing performance of the CEV model and compared the results with those by Bakshi *et al.* (1997). The CEV model, introducing only one more parameter compared with the Black–Scholes formula, improves the performance notably in all of the tests of in-sample, out-of-sample, and the stability of implied volatility. Furthermore, with a much simpler model, the CEV model can still perform better than the stochastic volatility model in short term and out-of-the-money categories. When applied to American option pricing, high-dimensional lattice models are prohibitively expensive. Our numerical experiments clearly show that the CEV model performs much better in terms of the speed of convergence to its closed-form solution, while the implementation cost of the stochastic volatility model is too high and practically infeasible for empirical work.

In summary, with a much less implementation cost and faster computational speed, the CEV option pricing model could be a better candidate than more complex option pricing models, especially when one wants to apply the

CEV process for pricing more complicated path-dependent options or credit risk models.

Keywords
Constant elasticity of variance (CEV) process, option pricing model, empirical performance, numerical experiment, stochastic volatility option pricing model, finite difference method of the SV model

Chapter 87: Asset allocation with cryptocurrencies

This research discusses the role of cryptocurrencies in portfolio investment and observes the timing within which the cryptos provide benefit to investors in a traditional financial market. We first use a mean-variance spanning test to check for any improvement that cryptos bring to a well-diversified portfolio and find a significant difference between portfolios with and without cryptos. Second, we analyze the weight dynamics of cryptos in the minimum-variance portfolio and the tangent portfolio to examine if cryptos present a hedging property in the mean-variance viewpoint. The finding shows that the optimal weights of cryptos increase distinctly in a market distress period, which shows their hedging property in a mean-variance view. Finally, we include cryptos in a well-diversified portfolio composed of common assets to check their weight dynamics in both tangent portfolio and minimum-variance portfolio. Consequently, we found that the cryptos take more weights in the tangent portfolio rather than in the minimum-variance portfolio, while the weights of cryptos increased in both portfolios during the COVID-19 pandemic; we thus conclude that cryptocurrencies can bring some hedging effect even in a portfolio with very common traditional assets. We also compare gold and cryptos and find that they have a similar pattern of weight dynamics, although gold has a slightly better effect in eliminating the downside risk of a minimum-variance portfolio.

Keywords
Cryptocurrencies, asset allocation, hedging

Chapter 88: Market-based, accounting-based, and composite-based beta forecasting

This chapter uses the concepts of basic portfolio analysis and the dominance principle to derive the CAPM. A graphical approach is first utilized to derive the CAPM, after which a mathematical approach to the derivation

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is developed that illustrates how the market model can be used to decompose total risk into two components. This is followed by a discussion of the importance of beta in security analysis and further exploration of the determination and the forecasting of beta. The discussion closes with the applications and implications of the CAPM, and the appendix offers empirical evidence of the risk–return relationship.

In this chapter, we define both market beta and accounting beta and how they are determined by different accounting and economic information. Then, we forecast both market beta and accounting beta. Finally, we propose a composite method to forecast beta.

Keywords
Total risk, systematic risk, non-systematic risk, market model, CAPM, accounting beta, market beta, composite forecasting

Chapter 89: Utility theory, capital asset allocation, and Markowitz portfolio selection model

In this chapter, we first discuss utility theory and utility function in detail, and then we show how asset allocation can be done in terms of quadratic utility functions. Based upon these concepts, we show Markowitz's portfolio selection model can be executed by constrained maximization approach. Real-world examples in terms of three securities are also demonstrated. In the Markowitz selection model, we consider that short sale is both allowed and not allowed.

Keywords
Asset allocation, concave utility function, indifference curve, Lagrangian objective function, linear utility function, risk aversion, short selling, utility theory, utility function

Chapter 90: Single-index model, multiple-index model, and portfolio selection

This chapter offers some simplifying assumptions that reduce the overall number of calculations of Markowitz models through the use of the Sharpe single-index and multiple-index models. Besides the single-index model, we also discuss how the multiple-index model can be applied to portfolio selection. We have theoretically demonstrated how single-index and multiple-index portfolio selection models can be used to replace the

Markowitz portfolio selection model. An Excel example of how to apply the single-index model approach is also demonstrated.

Keywords
Beta coefficient, covariance, Lagrangian calculus maximization, linear programming, market model, multiple-index model, single-index model

Chapter 91: Sharpe performance measure and Treynor performance measure approach to portfolio analysis

The main points of this chapter show how Markowitz's portfolio selection method can be simplified by either Sharpe performance measure or Treynor performance measure. These two approaches do not need to use constrained optimization procedure, however, these methods do require the existence of risk-free rate. Overall, this chapter has mathematically demonstrated how Sharpe measure and Treynor measure can be used to determine optimal portfolio weights.

Keywords
Lagrangian multipliers, risk-free rate, Sharpe performance measure, short sales allowed, short sales not allowed, Treynor performance measure

Chapter 92: Modelling different REIT cash flows

The accuracy and appropriateness of discounted cash flows (DCFs) have been debated for years in academia and industry. Broadly, most valuations, appraisers use accounting cash flows (ACF) in determining the present values (PVs) of firms. Some unique industries, such as the real estate investment trust (REITs), require customized DCFs to account for their capital structure among other factors. Fernández (2004, 2007) suggested that customized DCFs include debt cash flows, equity cash flows, free cash flows, and capital cash flows. The customized cash flows reveal different PVs for REIT firms. Furthermore, robustness results illustrate that customized DCFs are sensitive to selected macroeconomic (debt, equity, and funds from operations) and management (board and executive) variables. Fundamentally, the results are generalizable to the global REIT industry.

Keywords
Accounting cash flow, capital cash flow, debt cash flow, discounted cash flow, equity cash flow, free cash flow

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Chapter 93: Bayesian portfolio mean–variance efficiency test with Sharpe ratio's sampling error

This study proposes a Bayesian test for a test portfolio p's mean–variance efficiency that takes into account the sampling errors associated with the *ex post* Sharpe ratio $\hat{S}R$ of the test portfolio p. The test is based on the Bayes factor that compares the joint likelihoods under the null hypothesis H_0 and the alternative H_1, respectively. Using historical monthly return data of ten industrial portfolios and a test portfolio, namely, the CRSP value-weighted index, from January 1941 to December 1973 and January 1980 to December 2012, December 2012, the power function of the proposed Bayesian test is compared to the conditional multivariate F test by Gibbons *et al.* (1989) and the Bayesian test by Shanken (1987). In an independent simulation study, the performance of the proposed Bayesian test is also demonstrated.

Keywords
Bayesian test, Mean–variance efficiency, *ex post* Sharpe ratio, Bayes factor, CRSP value weighted index, conditional multivariate F test

Chapter 94: Fundamental analysis, technical analysis, and mutual fund performance

This chapter discusses methods and applications of fundamental analysis and technical analysis. In addition, it investigates the ranking performance of the value line and the timing and selectivity of mutual funds. A detailed investigation of technical versus fundamental analysis is first presented. This is followed by an analysis of regression time series and composite methods for forecasting security rates of return. Value line ranking methods and their performance then are discussed, leading finally into a study of the classification of mutual funds and the mutual fund managers' timing and selectivity ability. In addition, the hedging ability is also briefly discussed. Sharpe measure, Treynor measure, and Jensen measure are defined and analyzed. All of these topics can help improve performance in security analysis and portfolio management.

Keywords
Fundamental analysis, technical analysis, Dow theory, odd lot theory, confidence index, trading volume, moving average, component analysis, ARIMA models, composite forecasting, Sharpe performance measure, Treynor performance measure, Jensen performance measure

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Chapter 95: Synthetic options, portfolio insurance, and contingent immunization

This chapter discusses how futures, options, and futures options can be used in portfolio insurance (dynamic hedging). Four alternative portfolio insurance strategies are discussed in this chapter. These strategies are (i) stop-loss orders, (ii) portfolio insurance with listed put options, (iii) portfolio insurance with synthetic options, and (iv) portfolio insurance with dynamic hedging. In addition, the techniques of combining stocks and futures to derive synthetic options are explored in detail. Finally, important literature related to portfolio insurance is also reviewed.

Keywords
Synthetic option, stop-loss orders, put options, dynamic hedging, tail wag the dog, stock index futures

Chapter 96: Global international ELM versus momentum

We construct liquidity and earnings-based factors and combine them with the Market to describe stock returns. Liquidity and liquidity growth are significant factors across markets. Intercept tests show that the International Earnings, Liquidity, and Market (IELM) model fits the cross-section in various country groupings. As previous research showed a liquidity growth factor subsumes momentum in the U.S., we test this across international markets. From 2001 through 2019, the momentum factor has a high mean and is significant in Europe and in the Asia Pacific ex-Japan. For this time period, however, momentum is not significant in North America and Japan. While the IELM model reduces the momentum intercept in North America, both IELM and Fama and French (2017) have trouble explaining momentum in Europe and Asia where momentum is pervasive.

Keywords
Liquidity, momentum, earnings, asset pricing, factor models, GRS test, spanning tests

Chapter 97: Estimating European and American option pricing models: Excel and SAS language approach

The main purpose of this chapter is to show how Excel and SAS language can be used to estimate European options and American options. In this chapter, we use bivariate normal distribution to derive American options with one

dividend payment. Both Excel program and SAS program are presented in the appendices.

Keywords
Bivariate normal distribution, Black's model, Black–Scholes model, dividend yield, future option, stock indices

Chapter 98: Estimating the probabilities of default under the assumption of unobserved heterogeneity

Bond default and rank transition are often modeled as a Markov chain with an absorbing state. However, recent studies show that the theory does not match the empirical data. We suggest that this mismatch possibly arises from unobserved heterogeneity and we examine via a numerical example whether increased heterogeneity reduces, as expected, the accuracy of the estimated defaults and also the extent to which this reduction is economically significant. We then suggest a methodology for identifying the heterogeneity parameters and for testing their explanatory power.

Keywords
Bond rating, bond rating agencies, bond failure rates, transition probability matrix, probability of default, heterogeneity, homogeneity, Markov chains

Chapter 99: A factor model for graph data

Graph data have been becoming an important channel towards exploring relations among a large number of subjects in the big data era. In the past decades, community structures have been found in many complex real-world networks and play a key role in the modeling of graph data, for example, the stochastic block model and its extensions. However, recent studies unveil more sophisticated modules and typical examples include star and bipartite structures. In most graph models, these link-pattern modules are piled up in terms of multiple communities. This paper proposes a graph factor model in which each node is endowed with several (latent) features. Factors are channels for edge connections and can be characterized by link functions that map features of pairs of nodes to the edge probabilities. This model may naturally incorporate different kinds of link-pattern modules including communities, stars, and bipartite structures. The inference for the model can be carried out through an MCMC procedure. Both synthetic data and real-world networks are used for numerical illustrations.

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Keywords
Node clustering, community detection, link-pattern communities

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Chapter 100: Indices herding behavior and its impact on listed real estate and other two asset classes: A case of developed versus emerging markets

The literature on indices herding behavior among bonds, equities, and real estate is very thin. When one compares developed and emerging markets, specifically the United States, the United Kingdom, Taiwan, and South Africa, such studies are hard to find. This study uses the principal component analysis to extract and illustrate parameters' driving herding investment behavior for the indices of the mentioned countries. Thereafter, the vector autoregressive model is used for the robustness test. The results reveal, first, governmental relationships and similarities among countries influenced herding behavior in the selected capital markets indices. Second, most of the herding occurs in the bond indices for the four countries. Finally, the robustness results reveal spillover opportunities in between and across countries irrespective of the index analyses. The results are generalizable as they are consistent with prior studies such as Zaremba *et al.* (2021).

Keywords
Bonds, equities, herding, real estate

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Keywords
Bonds, equities, herding, real estate

Chapter 102: Price momentum, earnings forecasting, and valuation: Implications for inefficient markets

Financial anomalies have been studied in the U.S. However, recent evidence suggests that what were initially identified as return anomalies have diminished in U.S. data. Have regularities identified changed or are they persistent? Have historical and earnings forecasting data been a consistent and highly statistically significant, source of excess returns? We test a number of financial anomalies of the 1980–1990s and report that several models and strategies continue to produce statistically significant excess returns not absorbed by then-known factor models. We report that earnings forecasts, revisions, and breadth and price momentum have maintained their statistical significance during the May 1995–December 2017 time period. More importantly, we use expected return models and multifactor models estimated and known at the start of our current analysis, assuring our readers of out-of-sample and post-publication verification of the models.

Keywords
Expected returns, factor models, earnings forecast

Chapter 103: Advancement of optimal portfolio models with short-sales and transaction costs: Methodology and effectiveness

This paper presents the advancement of several widely applied portfolio models to ensure flexibility in their applications: mean–variance (MV), mean–absolute deviation (MAD), linearized value-at-risk (LVaR), conditional value-at-risk (CVaR), and omega models. We include short-sales and transaction costs in modeling portfolios and further investigate their effectiveness. Using the daily data of international ETFs over 15 years, we generate the results of the rebalancing portfolios. The empirical findings show that the MV, MAD, and omega models yield higher realized returns

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with lower portfolio diversity than the LVaR and CVaR models. The outperformance of these risk–return-based models over the downside risk-focused models comes from efficient asset allocation and not only the saving of transaction costs.

Keywords
Portfolio selection, conditional value-at-risk model, value-at-risk model, omega model, transaction costs, short selling

Chapter 104: Implied variance estimates for Black–Scholes and CEV OPM: Review and comparison

The main purpose of this chapter is to demonstrate how to estimate implied variance for both the Black–Scholes option pricing model (OPM) and constant elasticity of variance (CEV) OPM. For the Black–Scholes OPM model, we classify them into two different estimation routines: numerical search methods and closed-form derivation approaches. Both the MATLAB approach and approximation method are used to empirically estimate implied variance for American and Chinese options. For the CEV model, we present the theory and demonstrate how to use related Excel program in detail.

Keywords
Implied variance, Black–Scholes model, MATLAB approach, approximation approach, CEV model

Chapter 105: On the treatment of the momentum factor in accounting-based anomalies: A discussion

We survey the literature on anomalies in accounting research with a specific focus on whether and how it accounts for the momentum effect (Jegadeesh and Titman 1993, 2001). Even though accounting academics recognize the treatment of the momentum effect via inclusion in an extended Fama–French model to be appropriate, most extant empirical studies of accounting anomalies either do not account for the momentum effect or do so as a robustness check. Where included in the analysis, the momentum factor substantially reduces returns to portfolio strategies which exploit market underreaction. We argue that this treatment is in part due to the normal lag in the incorporation of research innovations but also likely due to persisting differences of opinion in the finance and accounting literature on whether to treat momentum as an anomaly or an asset pricing factor. More recent studies

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in accounting, however, seem to account for and treat the momentum effect more uniformly.

Keywords
Accounting anomalies, momentum factor, post-earnings announcement drift, return momentum effect

Chapter 106: Constant elasticity of variance option pricing model: Integration and detailed derivation

In this paper, we review the renowned constant elasticity of variance (CEV) option pricing model and give detailed derivations. There are two purposes of this article: First, we show the details of the formulae needed in deriving the option pricing and bridge the gaps in deriving the necessary formulae for the model. Second, we use a result by Feller to obtain the transition probability density function of the stock price at time T given its price at time t with. In addition, some computational considerations are given for the facilitation of computing the CEV option pricing formula.

Keywords
Constant elasticity of variance model, non-central chi-square distribution, option pricing

Chapter 107: Options and option strategies: Theory and empirical results

This chapter aims to establish a basic knowledge of options and the markets in which they are traded. It begins with the most common types of options, calls, and puts, explaining their general characteristics and discussing the institutions where they are traded. In addition, the concepts relevant to the new types of options on indexes and futures are introduced. The next focus is the basic pricing relationship between puts and calls, known as put–call parity. The final study concerns how options can be used as investment tools. Alternative option strategies theory has been presented. Excel is used to demonstrate how different option strategies can be executed.

Keywords
Put–call parity, long call, short call, long put, short put, long straddle, short straddle, long vertical (bull) spread, short vertical (bear) spread, calendar (time) spreads, long straddle, short straddle, long vertical (bull) spread, short vertical (bear) spread, protective put, covered call, collar

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Chapter 108: A cross-sectional asset pricing test with more power: An instrumental variable approach

I find that the downward bias of the estimated coefficient of betas in the Fama–MacBeth cross-sectional regression is caused by the endogeneity of the estimated betas due to measurement errors. I propose an instrumental variable methodology that purges the endogeneity. The purged betas have a 95% correlation with the original betas and retain the relation with firm size. I document that controlling for the purged betas in the Fama–MacBeth cross-sectional regression has higher statistical power to correctly reject the null hypothesis of the non-existence of size premium and also has higher R-squared and higher total sum of squares than the Brennan–Chordia–Subrahmanyam (1998) methodology.

Keywords
Fama–MacBeth regression, statistical power of test, endogeneity, bias of the estimated betas, instrumental variable, size variable

Chapter 109: Current vs. permanent earnings for estimating alternative dividend payment behavioral model: Theory, methods, and applications

Marsh and Merton (1987) and Garrett and Priestley (2000) have used aggregated permanent instead of current earnings to estimate aggregated dividend behavior models which were developed by Lintner (1956). Lee and Primeaux (1991) used permanent instead of current EPS to estimate Lintner's dividend payment behavior model for individual companies. Most recently, Lambrecht and Myer (2012) have theoretically shown that permanent, instead of current, EPS should be used to estimate the dividend payment behavior model for individual companies to avoid measurement error and misspecification of the model.

The main purposes of this paper are to (1) theoretically explain why firms generally allocate permanent earnings and transitory earnings between dividends payments and retained earnings, (2) develop alternative methods for decomposing current earnings into permanent and transitory components, (3) empirically estimate alternative dividend payment behavior models by using two alternative permanent EPS estimates for both individual firms and pooled data, and (4) test Lambrecht and Myer's (2012) theoretical results related to alternative dividend payment behavior models. We find that the average long-term payout ratio is downward biased and the average

estimated intercept is generally upward biased when the current instead of permanent EPS are used. We also find that the combined model performs well to deal with both measurement errors and specification errors in describing the dividend payment behavior model.

Keywords
Current earnings, current EPS, permanent earnings, permanent EPS, dividend behavior models, specification analysis, partial adjustment coefficient, long-term payout ratio

Chapter 110: Differential effect of inside debt, CEO compensation diversification, and firm investment

The main purposes of this paper are to study (1) a differential effect of inside debt on components of firm risk and (2) how it relates to CEO portfolio diversification to reduce firm risk exposure. We find that compensating CEOs with inside debt (e.g., pensions and other deferred compensation plans) leads to reductions in firms' systematic risk and idiosyncratic risk, to disproportionate degrees. CEOs with larger inside debt draft and implement policies, which lead to a significantly larger reduction in the idiosyncratic firm risk and investment. We then show that the differential effect is the result of an asymmetry in CEOs' perceived benefits of diversifying exposures to individual firm risk components. We further show that granting excessive debt-based pay may divert executives from firm-specific, but productive activities (e.g., R&D investments), therefore, may compromise long-run corporate success.

Keywords
Executive compensation, inside debt, risk-taking incentive, systematic and idiosyncratic risk, R&D investment, overconfidence

Chapter 111: Optimal payout ratio under uncertainty and the flexibility hypothesis: Theory, empirical evidence, and implications

Following the dividend flexibility hypothesis used by DeAngelo and DeAngelo (2006), Blau and Fuller (2008), and others, we theoretically extend the proposition of DeAngelo and DeAngelo's (2006) optimal payout policy in terms of the flexibility dividend hypothesis. We also introduce growth rate, systematic risk, and total risk variables into the theoretical model. In addition, based upon Lee and Alice (2021), we discuss the implication of

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the existence of optimal payout ratio in financial analysis and decision for a company.

To test the theoretical results derived in this chapter, we use data collected in the US from 1969 to 2009 to investigate the impact of growth rate, systematic risk, and total risk on the optimal payout ratio in terms of the fixed-effect model. We find that based on flexibility considerations, a company will reduce its payout when the growth rate increases. In addition, we find that a nonlinear relationship exists between the payout ratio and the risk. In other words, the relationship between the payout ratio and risk is negative (or positive) when the growth rate is higher (or lower) than the rate of return on total assets. Our theoretical model and empirical results can therefore be used to identify whether flexibility or the free cash flow hypothesis should be used to determine the dividend policy.

Keywords
Dividends, flexibility hypothesis, payout policy, fixed-effects model, financing policy, investment policy

Chapter 112: Sustainable growth rate, optimal growth rate, and optimal payout ratio: A joint optimization approach

A large number of studies have examined issues of dividend policy, while they rarely consider the investment decision and dividend policy jointly from a non-steady state to a steady state. We extend Higgins' (1977, 1981, and 2008) sustainable growth rate model and develops a dynamic model which jointly optimizes the growth rate and payout ratio. We optimize the firm value to obtain the optimal growth rate in terms of a logistic equation and find that the steady state growth rate can be used as the benchmark for the mean-reverting process of the optimal growth rate. We also investigate the specification error of the mean and variance of dividend per share when introducing the stochastic growth rate. Empirical results support the mean-reverting process of the growth rate and the importance of covariance between the profitability and the growth rate in determining dividend payout policy. In addition, the intertemporal behavior of the covariance may shed some light on the fact of disappearing dividends over decades.

Keywords
Dividend policy, payout ratio, growth rate, specification error, logistic equation, partial adjustment model, mean reverting process

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